AI 500
Foundations of Artificial Intelligence



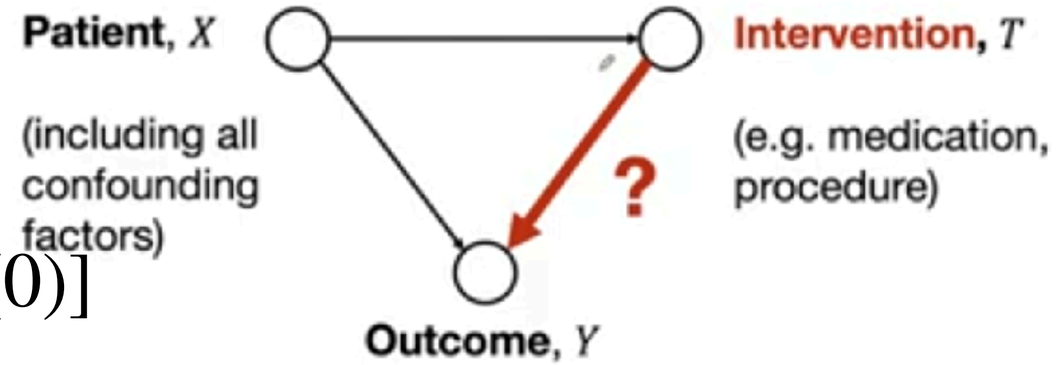
L10: Causal Inference II

By Murtaza Taj

LUMS
A Not-for-Profit University

Recap

- Given treatment (T) causes outcome (Y) if $E[Y(1)] \neq E[Y(0)]$
- Where $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$
- Individual causal effect $\tau_i = Y_i(1) - Y_i(0)$
- Average treatment effect $\mathbb{E}[\tau_i] = \mathbb{E}[Y_i(1) - Y_i(0)]$
- Is $\mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1) \mid T_i = 1] - \mathbb{E}[Y_j(0) \mid T_j = 0]$



Recap

Suppose we could observe both potential outcomes for all individuals!

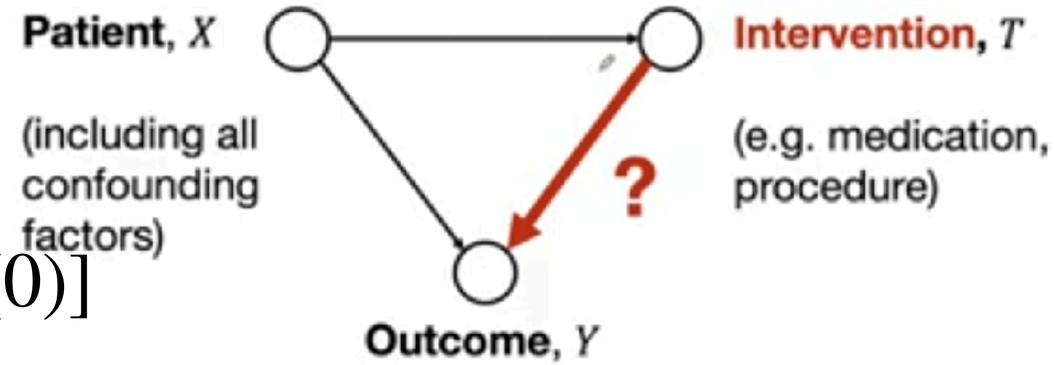
Person (i)	T_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	0.6	0.3	0.3
2	0	0.2	0.6	-0.4
3	1	0.4	0.2	0.2
4	0	0.3	0.1	0.2
5	1	0.7	0.5	0.2
6	0	0.6	0.4	0.2
7	0	0.3	0.1	0.2
		$\mathbb{E}[Y_i(1)] = 0.44$	$\mathbb{E}[Y_i(0)] = 0.31$	$\tau = 0.13$

$\mathbb{E}[Y_i(1)T = 1] = 0.57$ $\mathbb{E}[Y_i(0) \mid T = 0] = 0.3$ $D = 0.27$

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Recap

- Given treatment (X) causes outcome (Y) if $E[Y(1)] \neq E[Y(0)]$
- Where $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$
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- Is $\mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1) \mid T_i = 1] - \mathbb{E}[Y_j(0) \mid T_j = 0]$



- $D = \underbrace{\left(E[Y_i(1) \mid T = 1] - E[Y_i(0) \mid T = 1]\right)}_{\text{Effect of treatment on treated}} + \underbrace{\left(E[Y_i(0) \mid T = 1] - E[Y_i(0) \mid T = 0]\right)}_{\text{Selection bias}}$

Recap: Assumptions

Foundations of AI AI500

$$D = \underbrace{\big(E[Y_i(1) \mid T = 1] - E[Y_i(0) \mid T = 1]\big)}_{\text{Effect of treatment on treated}} + \underbrace{\big(E[Y_i(0) \mid T = 1] - E[Y_i(0) \mid T = 0]\big)}_{\text{Selection bias}}$$

- **AS1: Ignorability / Unconfoundedness** $(Y(0), Y(1)) \perp T \mid X$
- Treatment assignment is independent of potential outcomes given covariates

- **AS2: Overlap / Positivity** $0 < P(T = 1 \mid X) < 1$
- Every unit has a nonzero probability of receiving either treatment.

- **Stable Unit Treatment Value Assumption (SUTVA)**
- Each unit's outcome depends only on its own treatment, not others' treatments.

Causal Inference: Missing Data Problem



- Causal inference is the problem of estimating the counterfactual
  - Estimate missing data values using various methods
  - $Y_i(0)$ now becomes an estimated quantity, based on outcomes of other people who did not receive treatment

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Various Methodologies



- Randomized Control Trials (RCTs)
  - Gold standard

- Quasi-Experimental Methods
  - Examples: cross-sectional approaches, before-after comparison, matching, difference-in-differences, regression discontinuity, and instrumental variables
  - Learn causal structure/effects via flexible Machine Learning Based models

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Randomized Controlled Experiments

Randomized Controlled Experiments



- If we assign individuals to treatment and control at random, then the two groups are likely to be similar apart from the treatment
  - You can account – mathematically – for variability in the assignment

- Basic idea
  - Divide a population into treatment and control groups
  - Each group is selected at random to avoid systematic differences between individuals (or units)
  - Compare the average outcome across the two groups to find the average causal effect

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RCTs are the Gold Standard



- RCTs have long been a gold standard in the medical field, for example in establishing whether a new drug works

- They have also become increasingly common in other fields such as economics

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Caveats in RCTs



- Sometimes, people's responses in an experiment are influenced by their knowing which group they are in
  - So, you might want to run a blind experiment…
  - To make this work, you will have to give the control group a placebo, which is something…
- External validity (or generalizability) is always an important question

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But RCTs may not always be possible!



- RCTs can be expensive, infeasible, or unethical
  - For example, suppose you want to study the effects of smoking and you randomly assign some people to your “smoking” group

- What can we do when an experiment is not possible?
  - In such situations you will almost invariably be conducting an observational study, not an experiment
  - Be alert for confounding factors

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Observational Data

Let's now turn to the setting in which we don't have the ability to assign treatments randomly to units.

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A lot of observational data is out there

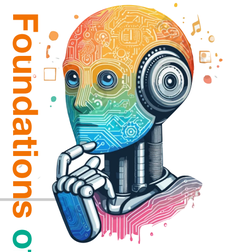


- Social media logs
- Credit card transactions
- Website/apps logs
- Satellite images
- …
- Can we do causal inference using observational data? What can go wrong?

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Unobserved Confounding

Is the New Drug Effective?



Goal: Evaluate the effectiveness of a new drug (treatment) in improving the recovery rate (outcome) from a disease

- T=0 → Control group (those who do not take the drug)
- T=1 → Treatment group (those who take the new drug)

We record the recovery rates of 700 patients who were given access to the drug. A total of 350 patients chose to take the drug and 350 patients did not.

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Is the New Drug Effective?



	Drug (T=1)	No Drug (T=0)	N
Men	93% (81/87)	87% (234/270)	357
Women	73% (192/263)	69% (55/80)	343
Combined Data	78% (273/350)	83% (289/350)	700

Is this new drug effective? Should we prescribe it to women? men? everyone?

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Is the New Drug Effective?



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The answer depends on the causal mechanism or the data generating process!

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Is the New Drug Effective?



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- Suppose we knew an additional fact: a women-only hormone has a negative effect on recovery, so women are less likely to recover than men, regardless of the drug.
- In addition, as we can see from the data, women are significantly more likely to take the drug than men are.

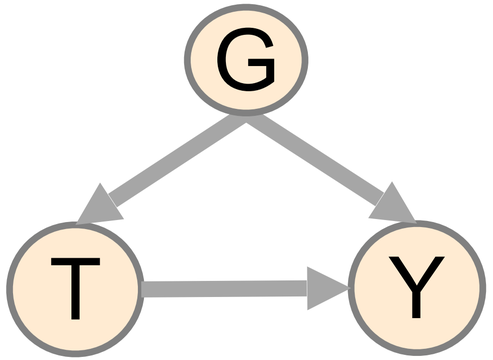
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Key Insight

- The reason the drug appears to be harmful overall is:

  – If we select a drug user at random, that person is more likely to be a woman and hence less likely to recover than a random person who does not take the drug

  – Put differently, being a woman is a common cause of both drug taking and failure to recover!



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Insight



- Thus, to assess the effectiveness of the drug, we need to compare subjects of the same gender

  - This will ensure that any difference in recovery rates can be attributed to the drug

  - Thus, we should consult the segregated data, which shows unequivocally that the drug is helpful

  - This matches our intuition, which tells us that the segregated data is “more specific,” hence more informative, than the unsegregated data

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Another Example

- Suppose we looked at the same numbers from our first example of drug taking and recovery…
  - … instead of recording participants' gender, patients' blood pressure were recorded at the end of the experiment

- In this case, we know that the drug affects recovery by lowering the blood pressure of those who take it—but unfortunately, it also has a toxic effect, which could influence the outcome
  - At the end of our experiment, we receive the results…

Results

	No Drug (T=0)	New Drug (T=1)	N
Low BP	93% (81/87)	87% (234/270)	357
High BP	73% (192/263)	69% (55/80)	343
Combined Data	78% (273/350)	83% (289/350)	700

Table 1. Results of a study into a new drug, with post-treatment blood pressure taken into account

Now, would you recommend the drug to a patient?

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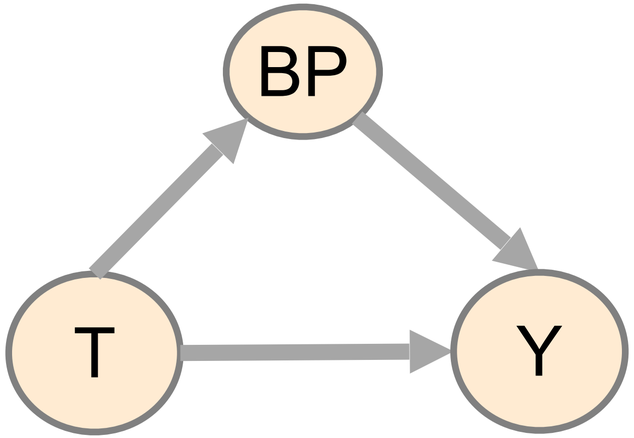
Correct Answer?

- Again, it depends on the data generating process!

  – In the general population, the drug might improve recovery rates because of its effect on blood pressure

  – But in the subpopulations—the group of people whose post-treatment BP is high and the group whose post-treatment BP is low—we, of course, would not see that effect; we would only see the drug's toxic effect

Why?



- Since lowering BP is one of the mechanisms by which treatment affects recovery, it makes no sense to separate the results based on blood pressure.

Decision



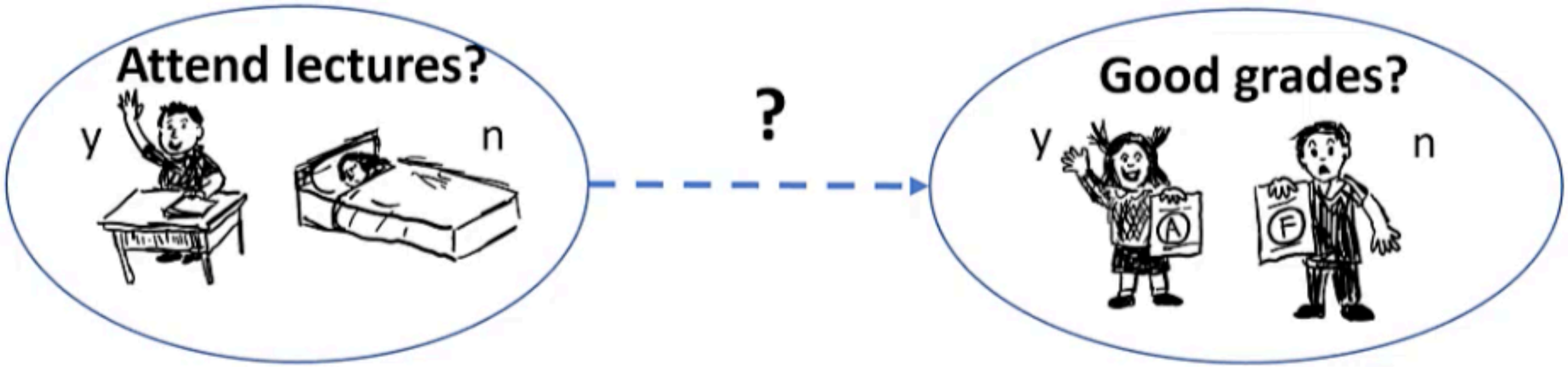
- In the low BP post-treatment group, some individuals may have responded well to the drug's BP-lowering effect
  - However, even with the drug working, toxic side effects might overshadow positive outcomes

- In the high BP post-treatment group, some people may not have responded to the drug, showing no benefit, while toxic effects become more apparent

- Looking at the general population, treatment increases the probability of recovery, leading us to recommend it

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Do students who skip lectures get better grades?



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We conduct an anonymous poll on LMS and find the following result (N=30)



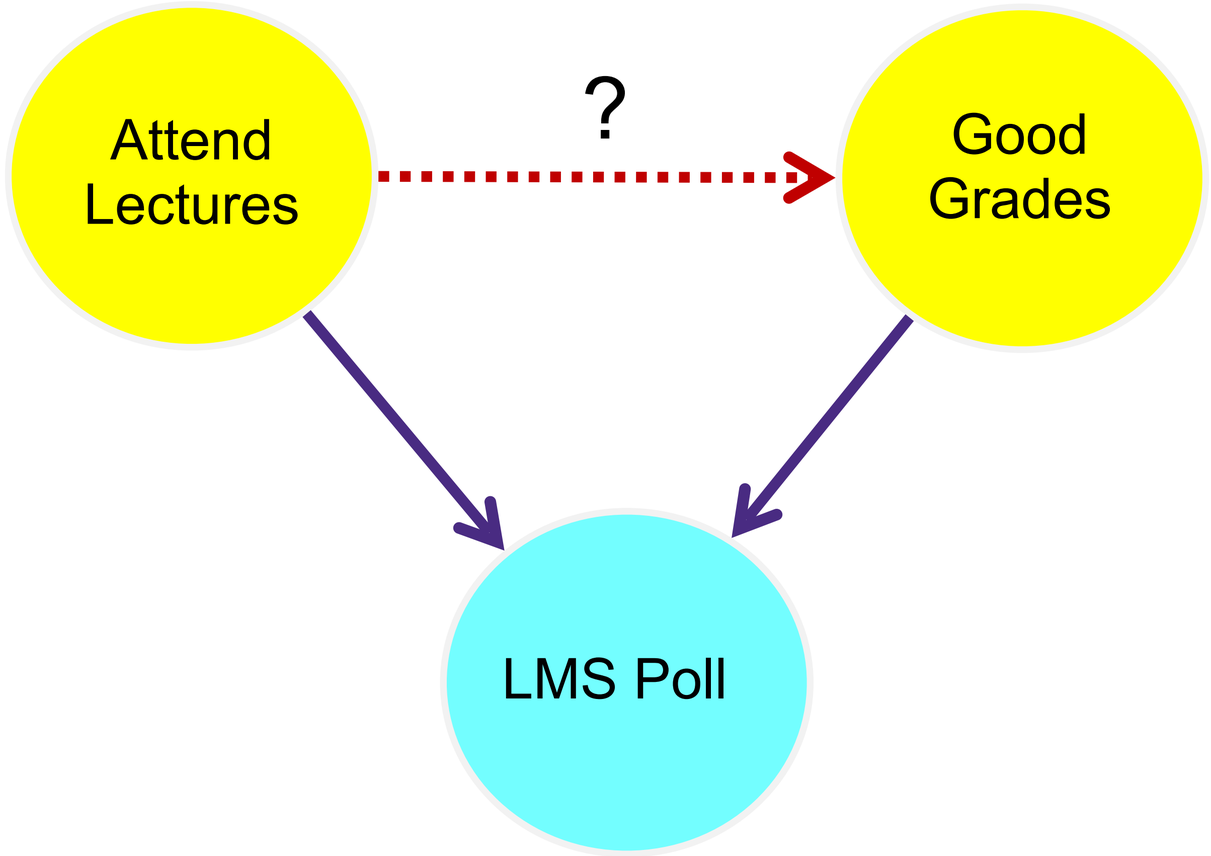
	% Getting Good Grades	Good Grades	Bad Grades
Attends class	72.7%	16	6
Does **not** attend class	75%	6	2

Estimated ATE = 75 – 72.7 = 2.3%

(*Not attending class leads to improvement in grades*!)

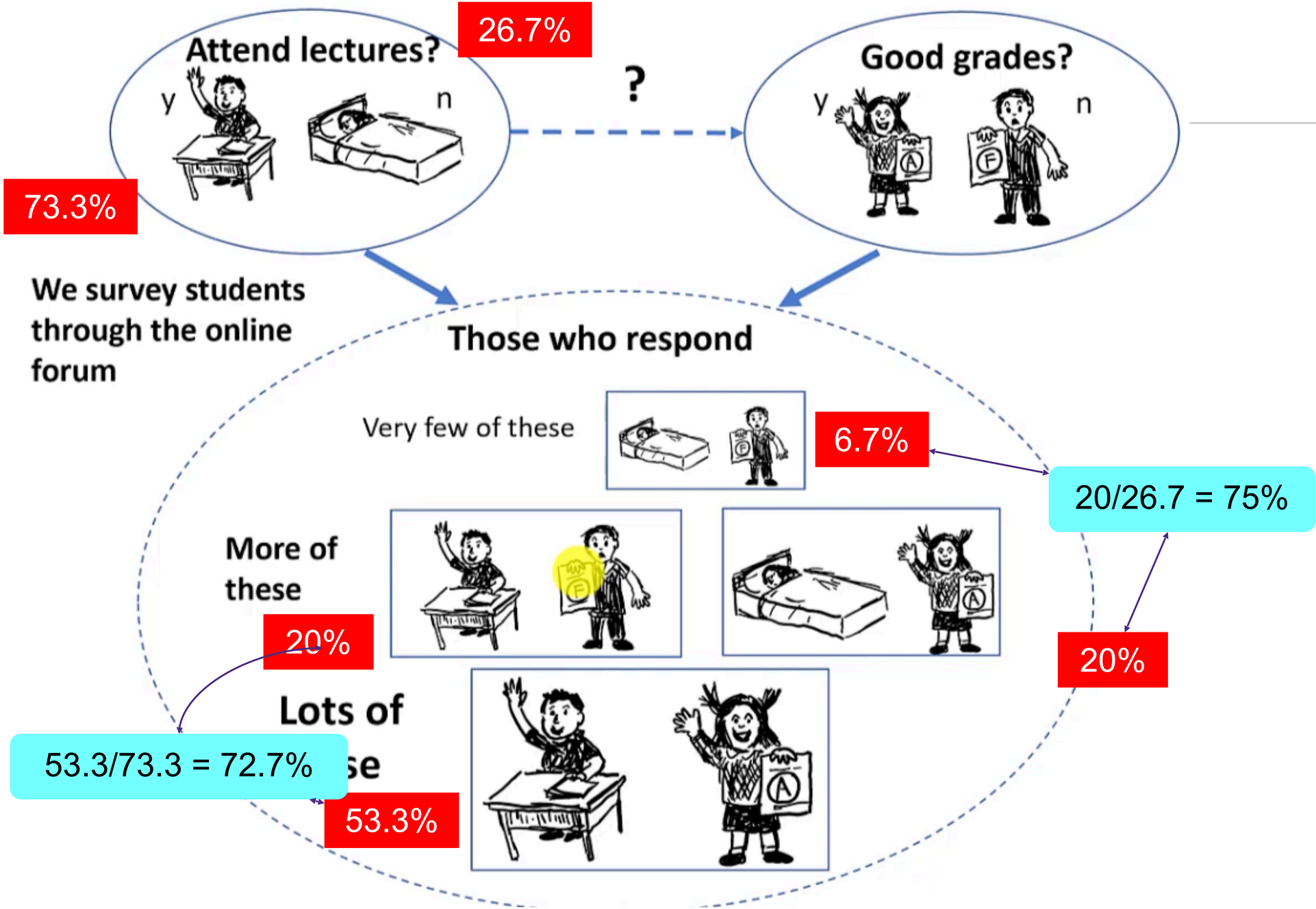
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Berkson's Paradox



This is known as Berkson's Paradox!

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Data from an Experiment

	% Getting Good Grades	Good Grades	Bad Grades
Attends class	62.5%	20	12
Does **not** attend class	37.5%	12	20

Estimated ATE = 62.5 – 37.5 = 25%

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Non-Experimental or Quasi-Experimental Methods

Types of Impact Evaluation Methods



- Non-Experimental or Quasi-Experimental Methods
  - Pre-Post
  - Cross-Sectional Differences
  - Differences-in-Differences
  - Conditioning on Variables
  - Multivariate Regression
  - Regression Discontinuity Design (RDD)
  - Instrumental Variables (IV)
  - Matching, Weighting, Synthetic Control, etc

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Pre-post (before-and-after)



- Comparison of outcomes of the **same group** before and after participating in the program (a.k.a. within subject design)
- Who is the comparison group?
  - Participants **prior to the start** of the program
- Advantage: Relatively simple
- Assumption(s) required
  - **No changes in the environment or individuals over time**

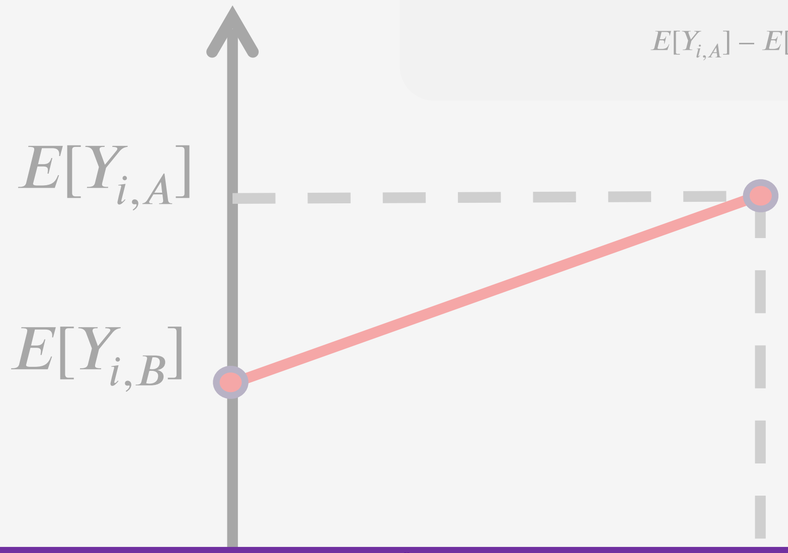
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Pre-post: Example

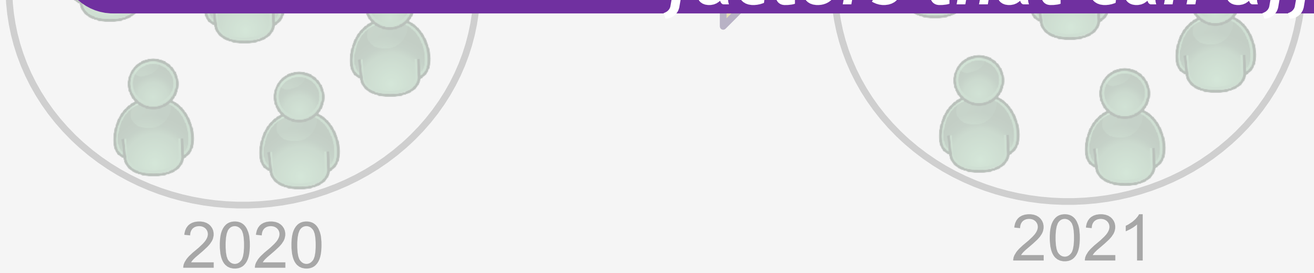
- **Intervention**: Free vocational training program
- **Outcome**: Income

Avg. Causal Effect =

$$E[Y_{i,A}] - E[Y_{i,B}]$$

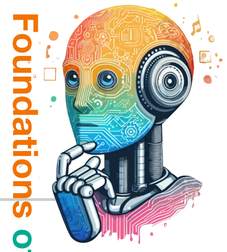


A trainee's income might have increased even if she had not taken the training—because of her own efforts, because of changing labor market conditions, or because of many other factors that can affect income!



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Pre-Post



- Pre-post design is most useful in demonstrating the **immediate impacts** of short-term interventions

- It is less useful for evaluating **longer term interventions**
  - Over a longer period of time, more circumstances can arise that may obscure the effects of an intervention

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Cross-Sectional Comparisons

- Comparisons across individuals at the **same time** (i.e., who received treatment and those who did not)

- Who makes up the comparison group? **non-participants**

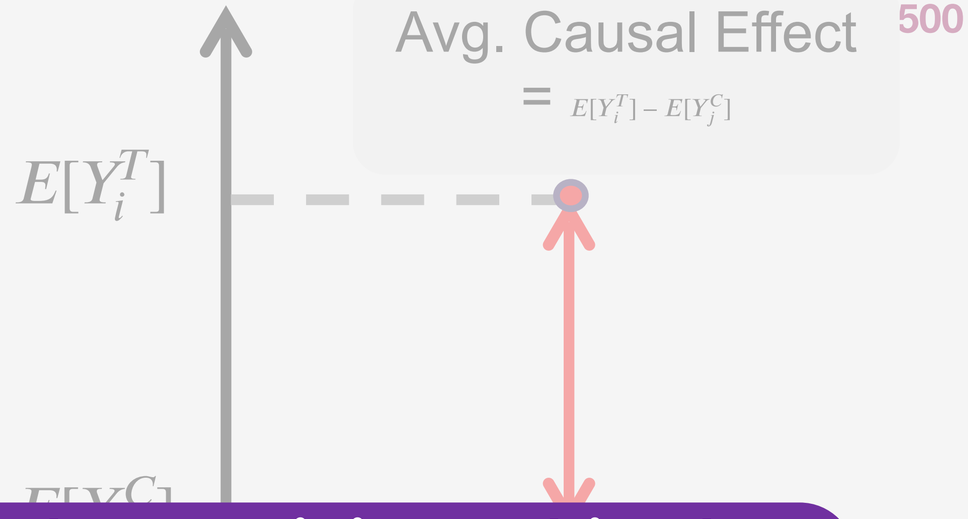
- Advantage(s):
  - Can be undertaken after the program has started

- Assumption(s) required
  - Participants are **not different** from non-participants in any way, including in the potential for future change

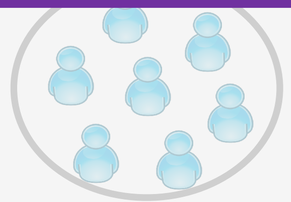
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Cross-Sectional Comparisons

- **Intervention**: A microfinance program that offers small loans
- **Outcome**: Income



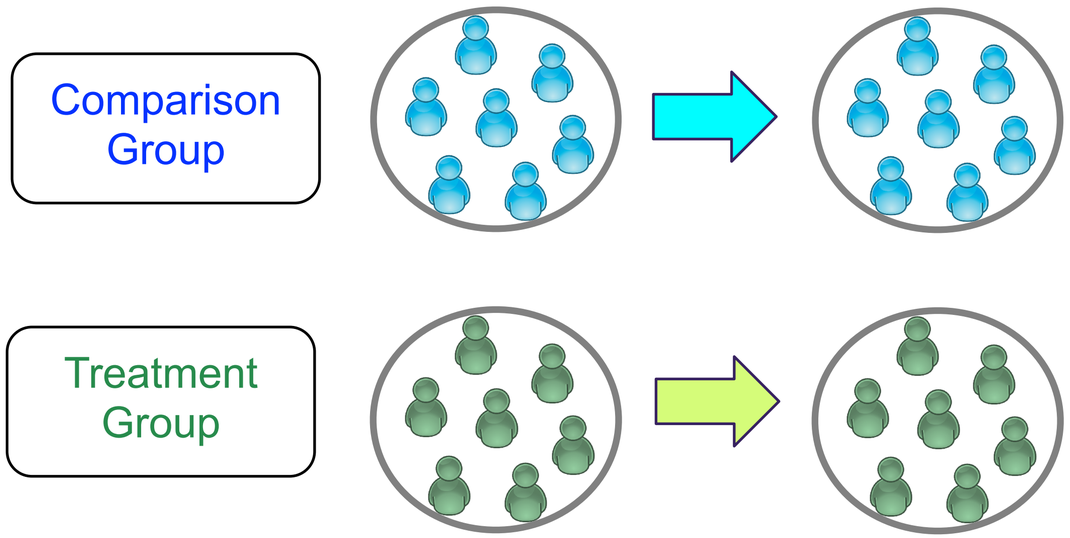
Major Challenge: Selection bias! those who participated in the program may have different characteristics from those who did not (e.g., increase in income may be due to the other characteristics)

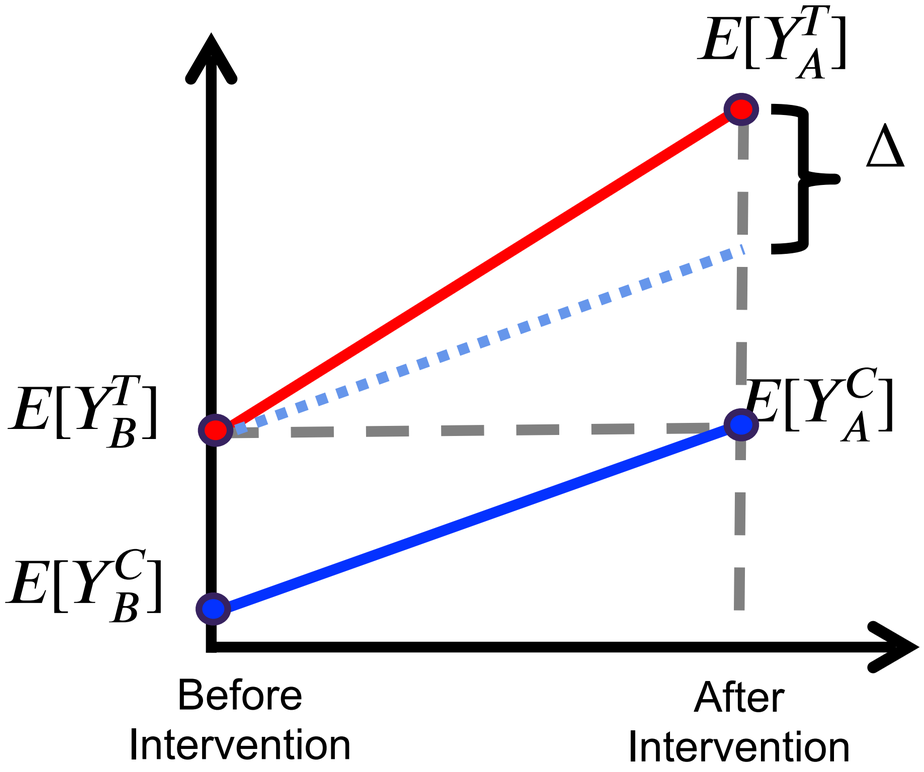


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Difference-in-Differences: Example

- **Intervention**: A road repair program at district-level
- **Outcome**: Employment rate due to improved access to labor markets





Avg. Causal Effect $(\Delta) = (E[Y_A^T] - E[Y_A^C]) - (E[Y_B^T] - E[Y_B^C])$

Note: We drop the index i from the expressions for clarity

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Difference-in-Differences (DiD)



- Compares **changes over time** between participants & non-participants
- Advantages
  - Accounts for time-varying changes **common to both groups**
- Assumption(s) required
  - The **only differences** between participants and non-participants are in the levels of observed and unobserved outcomes **at the start**
  - If the program had not existed, the participants and non-participants would have experienced identical trajectories during the study period (**parallel trends assumption**)

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Causal Inference with Machine Learning

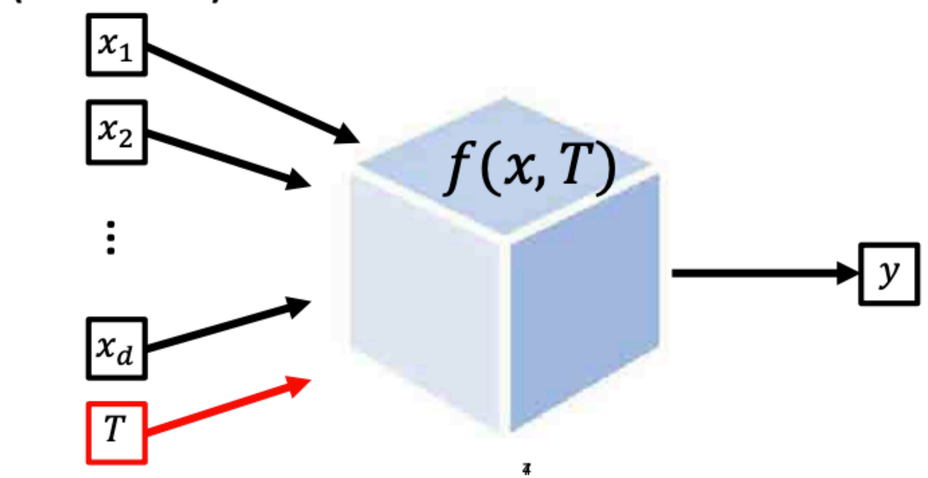
Common approaches for counterfactual inference

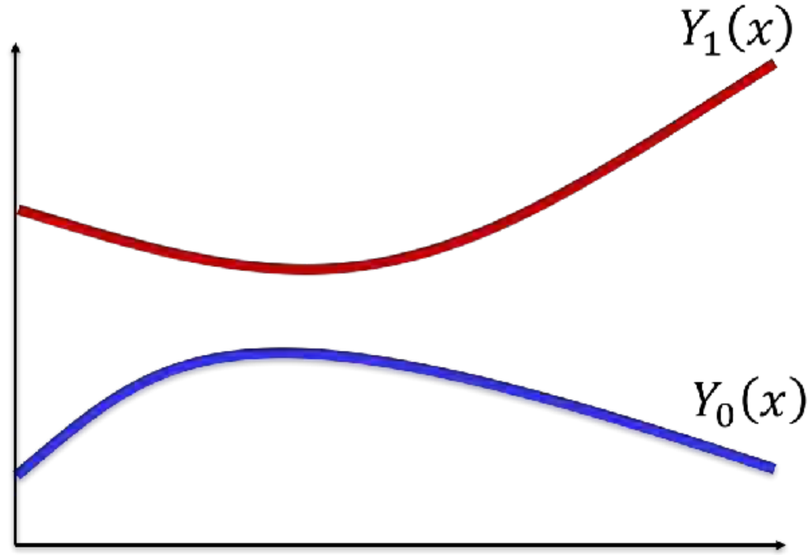


- Covariate adjustment
- Multi-variate Regression
- Propensity scores
- Matching

Covariate adjustment

- Explicitly model the relationship between treatment, confounders, and outcome
- Also called “Response Surface Modeling”





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Covariate adjustment

- Explicitly model the relationship between treatment, confounders, and outcome

- Under ignorability, the expected causal effect of T on Y:

$$CATE(x) = \mathbb{E}_{x \sim p(x)} \left[\mathbb{E}[Y_1 \mid T = 1, x] - \mathbb{E}[Y_0 \mid T = 0, x] \right]$$

- Fit a model $f(x, t) \approx \mathbb{E}[Y_t \mid T = t, x]$, then:

$$\widehat{CATE}(x_i) = f(x_i, 1) - f(x_i, 0)$$

$$\widehat{ATE}(x_i) = \frac{1}{n} \sum_{i=1}^{n} [f(x_i, 1) - f(x_i, 0)]$$

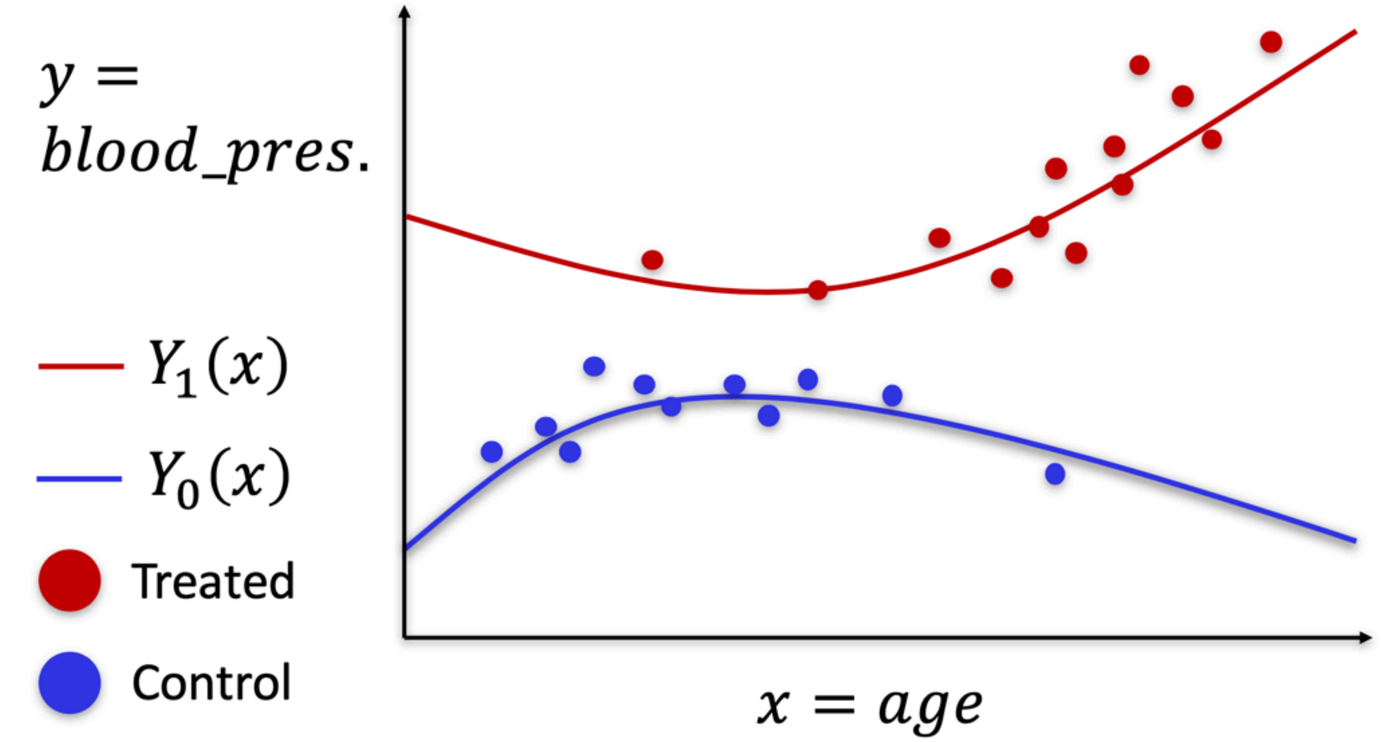
Covariate adjustment

$$\widehat{ATE}(x_i) = \frac{1}{n}\sum_{i=1}^{n}[f(x_i,1) - f(x_i,0)]$$

- Where is the observed data in here?
- What are our assumptions here?

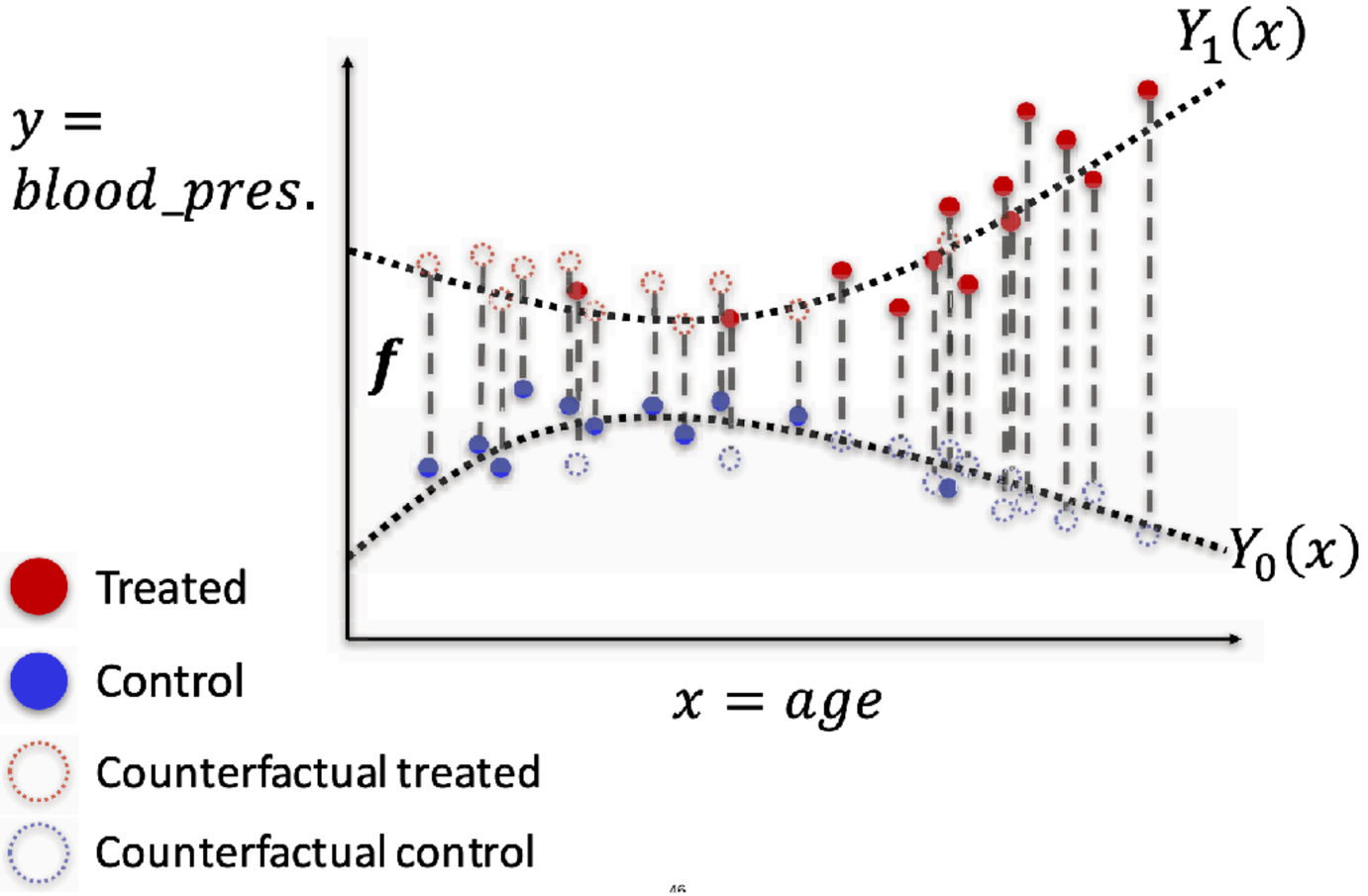
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Covariate adjustment



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Covariate adjustment



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Covariate adjustment

How would you actually know that your ML algo $f(\ .\ )$ has learned the relationship between input X and the treatment T and the outcome Y

- How do I know that the function $f(\ .\ )$ does a good job to learn the average treatment effect, particularly in high dimensional data.

$$\widehat{ATE}(x_i) = \frac{1}{n}\sum_{i=1}^{n}[f(x_i,1) - f(x_i,0)]$$

- Your algo might never learn the actual dependence on T, it might just through away T and learn the dependence of Y on X

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Covariate adjustment

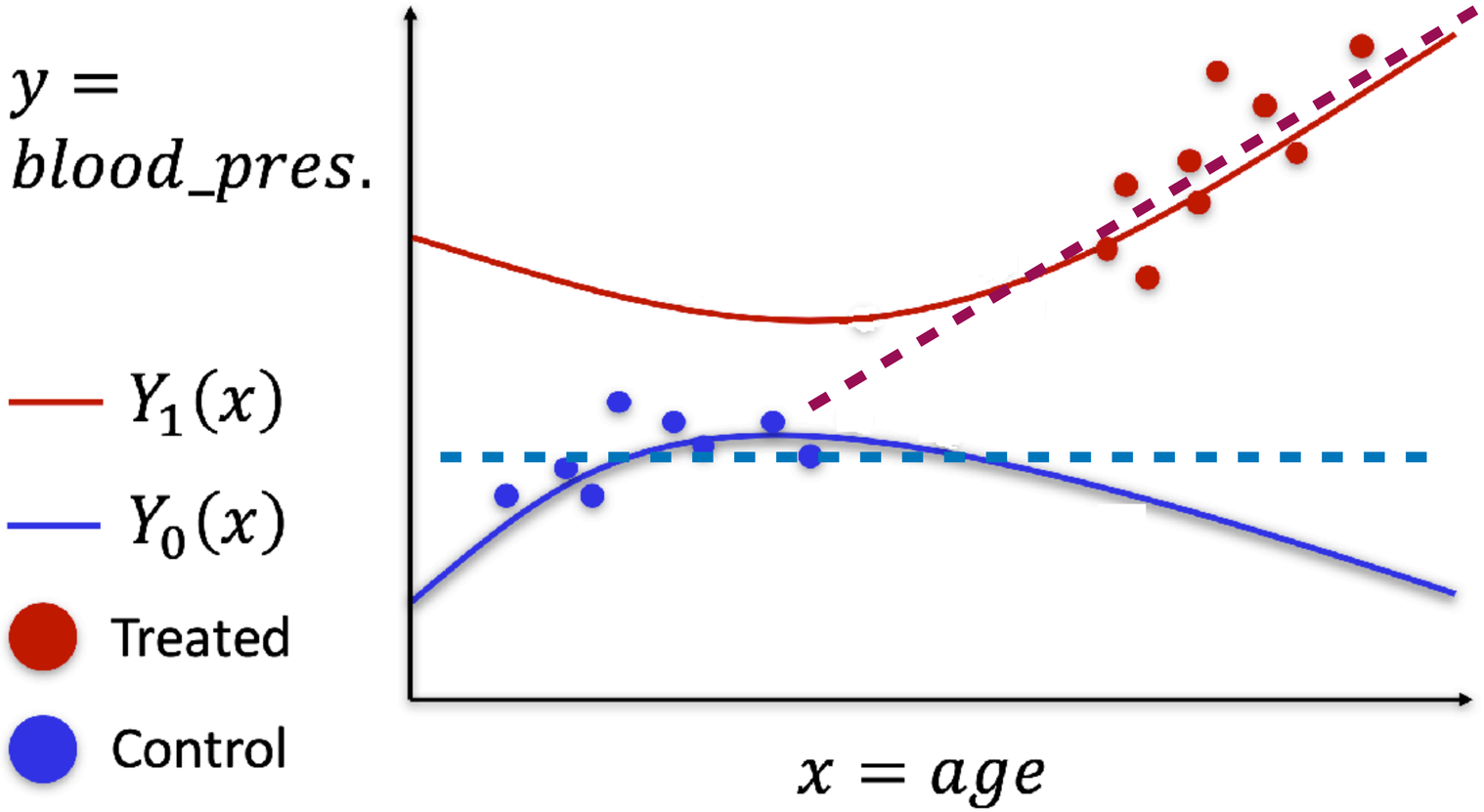
- Where does the overlap assumption show up in our approach for estimating ATE using covarint adjustment?

$$\widehat{ATE}(x_i) = \frac{1}{n}\sum_{i=1}^{n}[f(x_i,1) - f(x_i,0)]$$

- If you have overlap, then as the amount of data goes to infinity then you will eventually extrapolate correctly.

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How covariate adjustment fails when there is no overlap



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Covariate adjustment with linear models

- Assume that:

Blood pressure age medication

$$Y_t(x) = \beta x + \gamma \cdot t + \epsilon_t$$
$$\mathbb{E}[\epsilon_t] = 0$$

- Single covariate i.e. age
- Then:

$$CATE(x) := \mathbb{E}[Y_1(x) - Y_0(x)] =$$

$$\mathbb{E}[(\beta x + \gamma + \epsilon_1) - (\beta x + \epsilon_0)] = \gamma$$

$$ATE := \mathbb{E}_{p(x)}[CATE(x)] = \gamma$$

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Covariate adjustment with linear models

- Assume that:

Blood pressure age medication

$$Y_t(x) = \beta x + \gamma \cdot t + \epsilon_t$$
$$\mathbb{E}[\epsilon_t] = 0$$

$$ATE := \mathbb{E}_{p(x)}[CATE(x)] = \gamma$$

- For causal inference, need to estimate γ well, not $Y_t(x)$ - identification, not prediction
- γ tells you giving the treatment vs not giviging the treatment (for treatment increases or decreases the blood pressure)

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Covariate adjustment with non-linear models



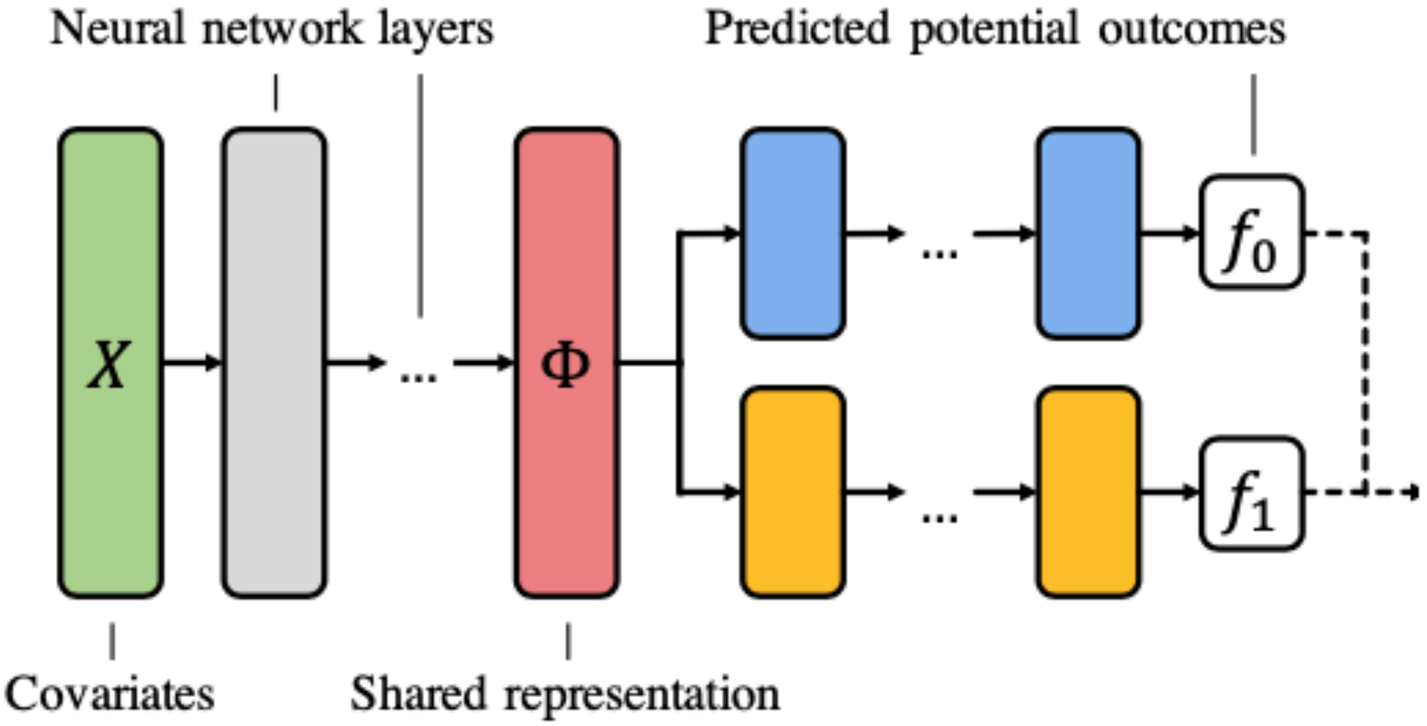
- Random forests and Bayesian trees
  Hill (2011), Athey & Imbens (2015), Wager & Athey (2015)

- Gaussian processes
  Hoyer et al. (2009), Zigler et al. (2012)

- Neural networks
  Beck et al. (2000), Johansson et al. (2016), Shalit et al. (2016), Lopez-Paz et al. (2016)

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Covariate adjustment with non-linear models



Shalit, Johansson, Sontag. *Estimating Individual Treatment Effect: Generalization Bounds and Algorithms*. ICML, 2017

Covariate Adjustment: Matching

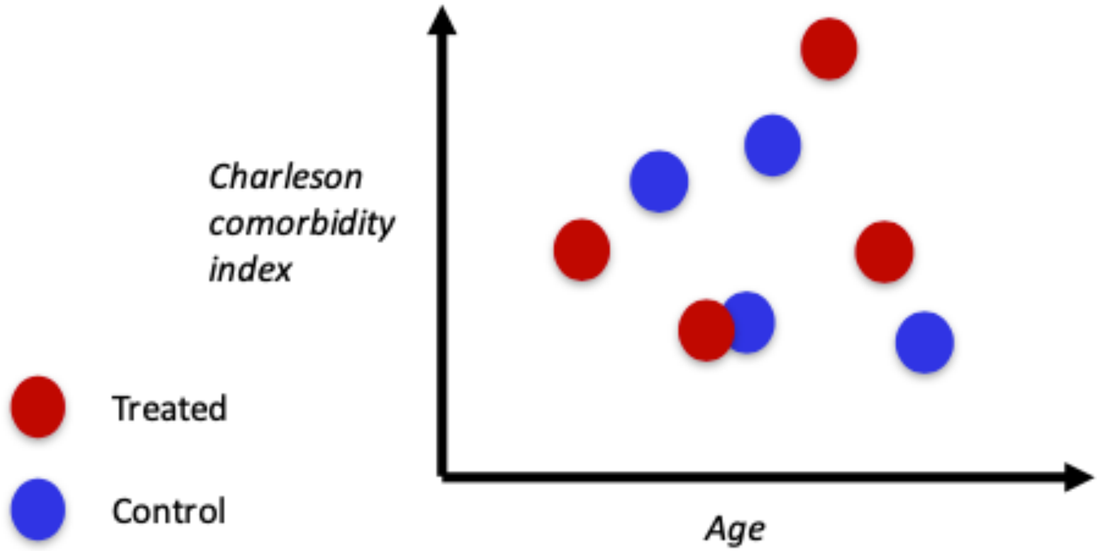


- Find each unit's long-lost counterfactual identical twin, check up on his outcome
- Used for estimating both ATE and CATE

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Match to nearest neighbor from opposite group

- You want to find a very similar sample but that didn't get the same treatment and we analyse of the outcome is same or changed?

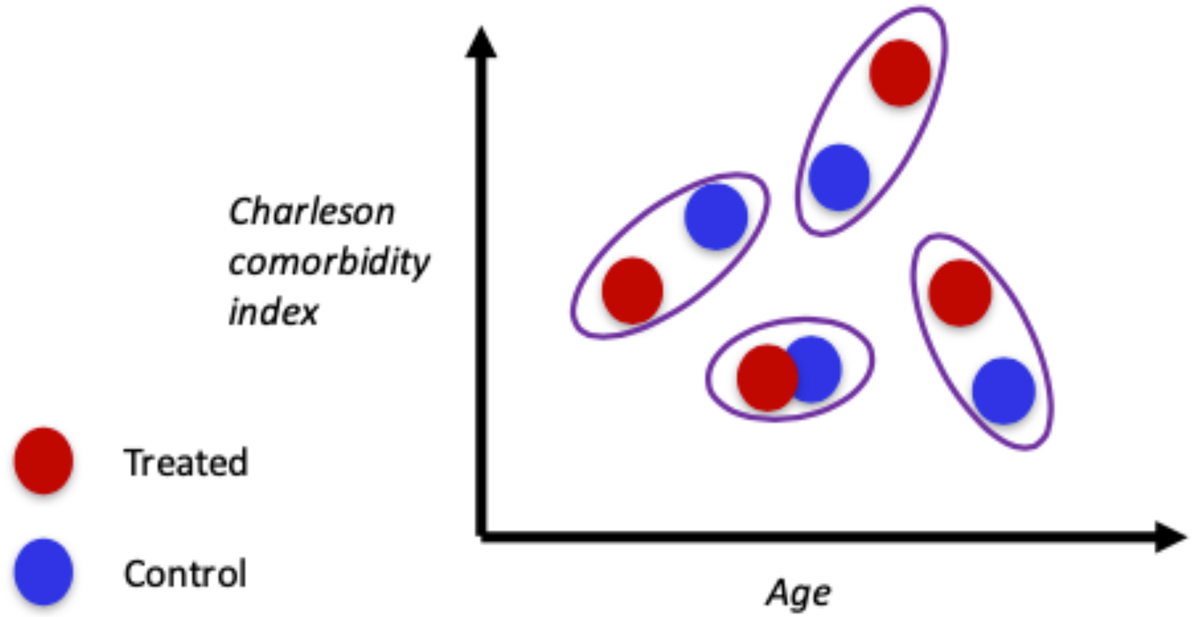


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Match to nearest neighbor from opposite group



- You want to find a very similar sample but that didn't get the same treatment and we analyse the outcome is same or changed?



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1-NN Matching

- Let $d(\,.\,,.\,)$ be a matric between x's
- For each i, define $j(i) = \arg \min_{j \ s.t. \ t_j \neq t_i} d(x_j, x_i)$. $j(i)$ is the nearest counterfactual neighbour of i
- $t_i = 1$, unit i is treated:

$$\widehat{CATE}(x_i) = y_i - y_{j(i)}$$

- $t_i = 0$, unit i is control:

$$\widehat{CATE}(x_i) = y_{j(i)} - y_i$$

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1-NN Matching

- Let $d(\,.\,,.\,)$ be a matric between x's
- For each i, define $j(i) = \arg\min_{j \ s.t. \ t_j \neq t_i} d(x_j, x_i)$. $j(i)$ is the nearest counterfactual neighbour of i

$$\widehat{CATE}(x_i) = (2t_i - 1)(y_i - y_{j(i)})$$

$$\widehat{ATE}(x_i) = \frac{1}{n}\sum_{i=1}^{n} \widehat{CATE}(x_i)$$

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1-NN Matching



- Interpretable, especially in small-sample regime
- Nonparametric
- Heavily reliant on the underlying metric (distance function)
- Could be misled by features which don’t affect the outcome

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Propensity Score

- Tool for estimating ATE

- Image that we had a data from a randomized control trial (RCT). Then we could simply estimate the ATE using:

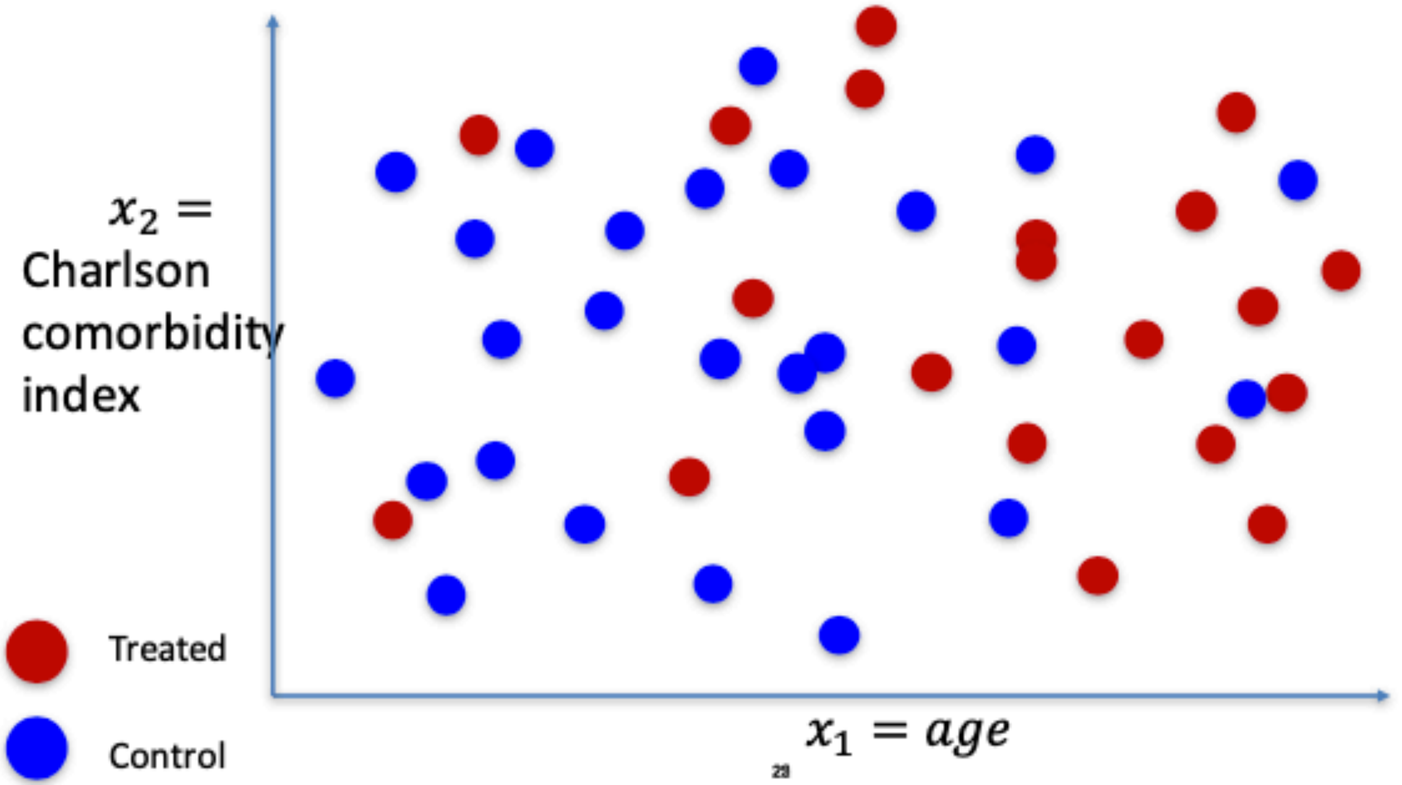
$$\frac{1}{n_1} \sum_{i \ s.t. \ T_i=1} Y_i - \frac{1}{n_0} \sum_{i \ s.t. \ T_i=0} Y_i$$

- Basic idea: turn observational study into a pseudo-randomized trial by re-weighting samples, similar to importance sampling

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Inverse propensity score re-weighting

$$p(x|t=0) \neq p(x|t=1)$$
$$\underset{control}{} \qquad \underset{treated}{}$$

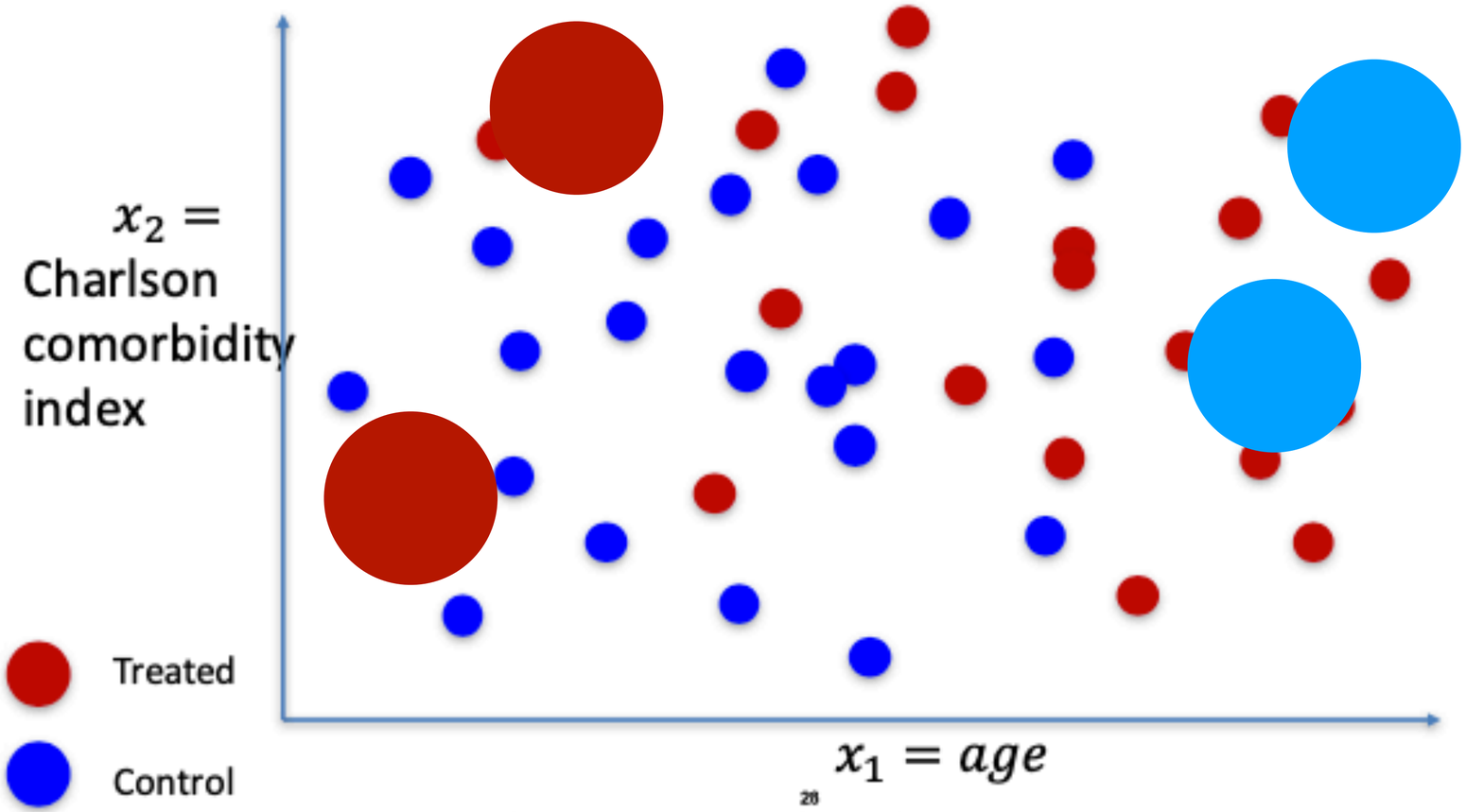


Young people received blue treatment whreas older people received red treatment

Inverse propensity score re-weighting

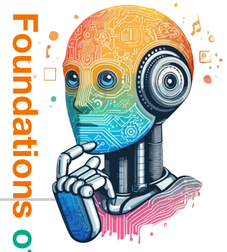
$$p(x|t=0) \cdot w_0(x) \approx p(x|t=1) \cdot w_1(x)$$

reweighted control *reweighted treated*



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Propensity score



- Propensity score: $p(t = 1,x)$, using ML tools, here we are ignoring Y
- Here we are predicting T given X
- Samples re-weighted by the inverse propensity score of the treatment they received

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Propensity scores – algorithm

- **Inverse probability of treatment weighted estimator**
- How to calculate ATE with propensity score for sample $(x_1, t_1, y_1), \cdots, (x_n, t_n, y_n)$

- Use any ML method to estimate $\hat{p}(T = t \,|\, x)$

$$\hat{ATE} = \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{\hat{p}(t_i = 1|x_i)} - \frac{1}{n} \sum_{i \text{ s.t. } t_i=0} \frac{y_i}{\hat{p}(t_i = 0|x_i)}$$

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Propensity scores – algorithm

- **Inverse probability of treatment weighted estimator**
- How to calculate ATE with propensity score for sample $(x_1, t_1, y_1), \cdots, (x_n, t_n, y_n)$

- Randomized trail $\hat{p}(T = t \,|\, x) = 0.5$

$$\hat{ATE} = \frac{1}{n} \sum_{i \text{ s.t. } t_i=1} \frac{y_i}{0.5} - \frac{1}{n} \sum_{i \text{ s.t. } t_i=0} \frac{y_i}{0.5} =$$

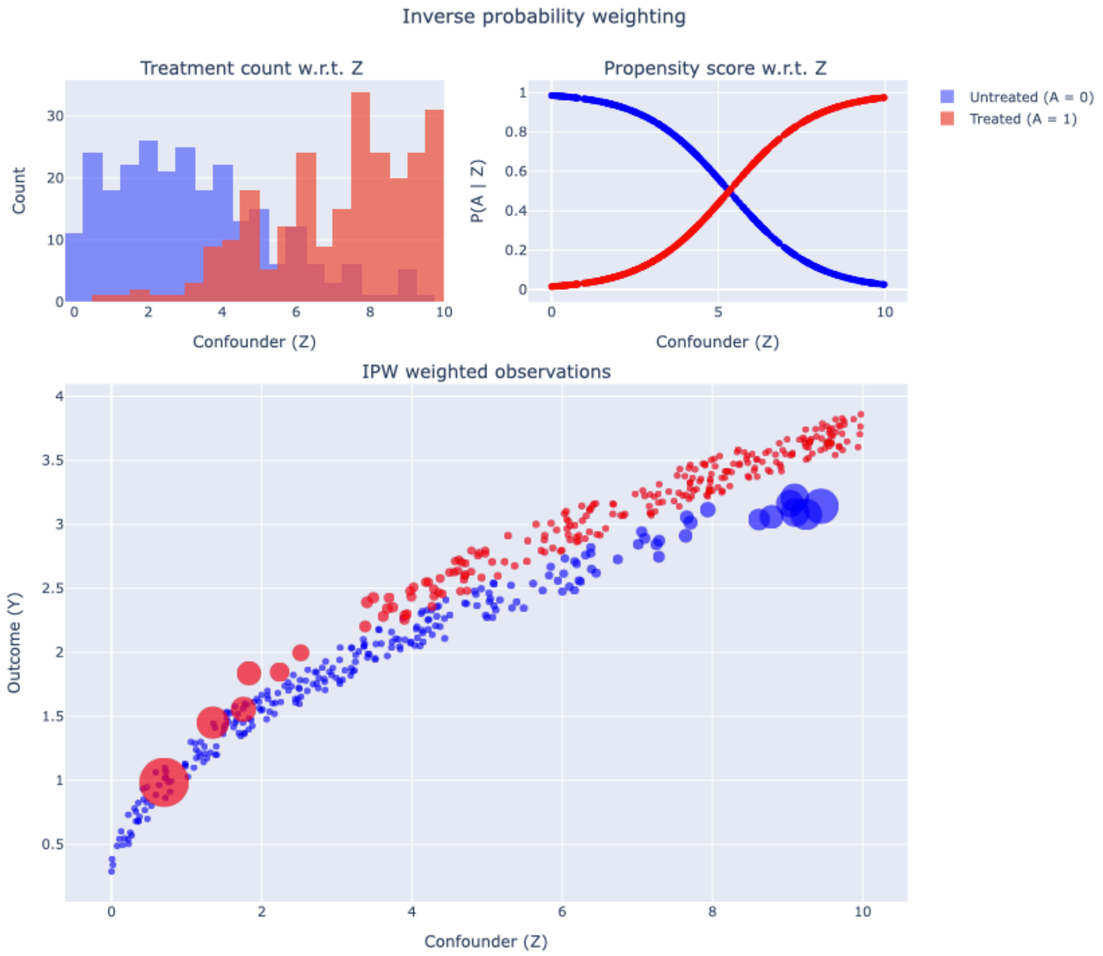
$$\frac{2}{n} \sum_{i \text{ s.t. } t_i=1} y_i - \frac{2}{n} \sum_{i \text{ s.t. } t_i=0} y_i$$

Sum over ~$\frac{n}{2}$ terms

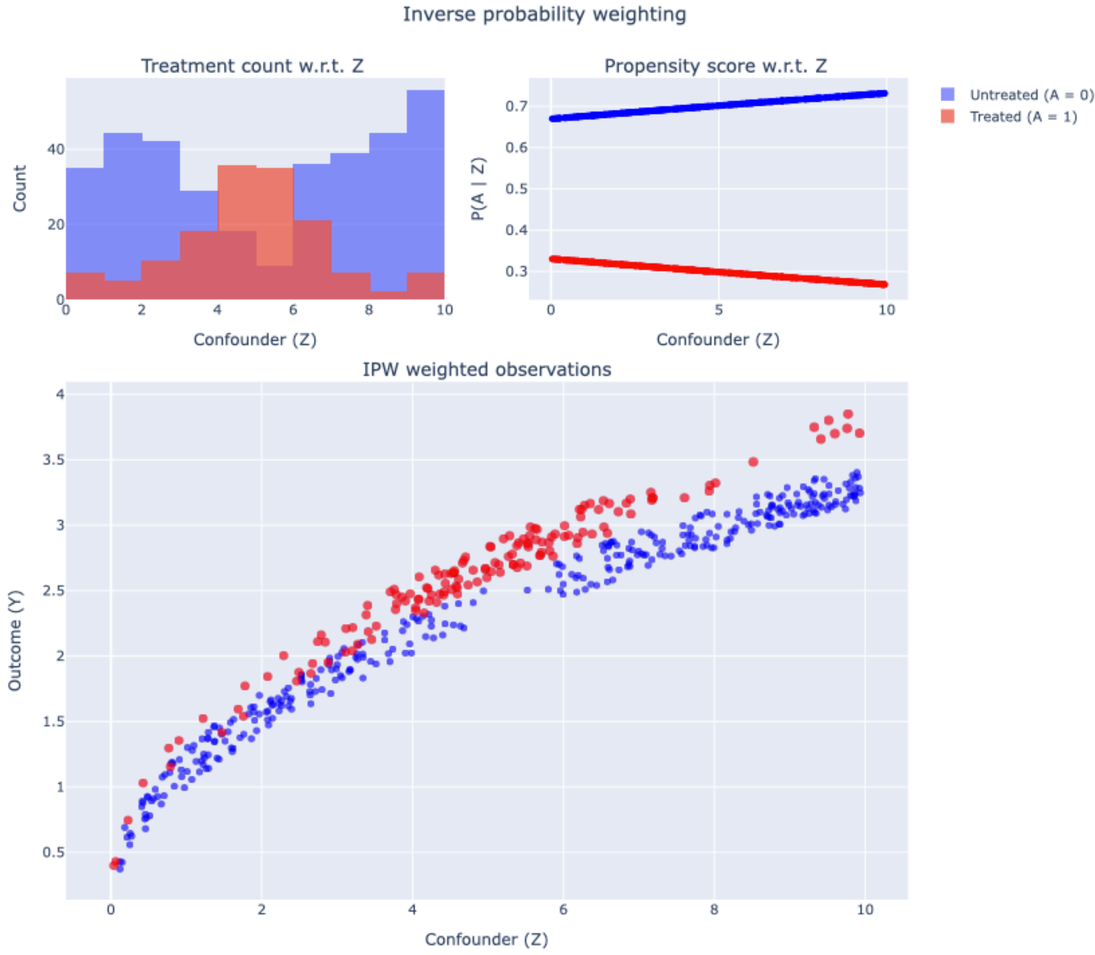
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Exercise 4: Inverse probability weigthing

- https://colab.research.google.com/drive/13Bsvvl5l3uR1hbVdpMAFR13gdjwoJ6if?usp=sharing#scrollTo=lF6gPBNuK9Re



Increase with z



Center on z

Problems with Inverse Propensity Weighting



- Need to estimate propensity score (problem in all propensity score methods)

- If there's not much overlap, propensity scores become non-informative and easily miscalibrated

- Weighting by inverse can create large variance and large errors for small propensity scores
  – Exacerbated when more than two treatment

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Propensity Weighting - Example



- Goal: Estimate the causal effect of a training program (T) on income (Y), where people with higher education (X) are much more likely to join the program.

Summary



- Two approaches to use machine learning for causal inference
  - Predict outcome given features and treatment, then use resulting model to impute counterfactuals (covariate adjustment)
  - Predict treatment using features (propensity score), then use to reweight outcome or stratify the data

- Consistency of estimates depend on:
  - Causal graph being correct (i.e., no unobserved)
  - Identifiability of causal effect (i.e., overlap)
  - Nonparametric regression is used (or correctly specified model)

Case Study: Balsakhi Program

Example: Balsakhi Program



Balsakhi Program: Background

- Problem:
  - Many children in 3rd and 4th standard were not even at the 1st standard level of competency
  - Class sizes were large
  - Social distance between teacher and many of the students was large

- Proposed solution:
  - Hire local women (balsakhis) from the community and train them to teach basic competencies (reading, numeracy) to lowest performing students
  - Implemented by Pratham, an NGO from India
  - In Vadodara, the balsakhi program was run in government primary schools in 2002-2003
  - Teachers decided which children would get the balsakhi

Balsakhi: Outcomes

- Children were tested at the beginning of the school year (Pretest) and at the end of the year (Post-test)

- QUESTION: How can we estimate the impact of the balsakhi program on test scores?

Randomized Evaluation

- Suppose we evaluated the balsakhi program using a randomized evaluation

- QUESTION #1: What would this entail? How would we do it?

- QUESTION #2: What would be the advantage of using this method to evaluate the impact of the balsakhi program?

The basics

- Take a sample of program applicants
- Randomly assign them to either:
  - Treatment Group – is offered the program
  - Control Group – not allowed to receive the program (during the evaluation period)
- The two groups will, on average, have the same observable and unobservable characteristics
  - since assignment is purely by chance
  - provided we have a large enough number of units
- Impact = Difference in outcomes between the treatment and control groups after the program

Key advantage of experiments

Because members of the groups (treatment and control) do not differ systematically at the outset of the experiment,

any difference that subsequently arises between them can be attributed to the program rather than to other factors.

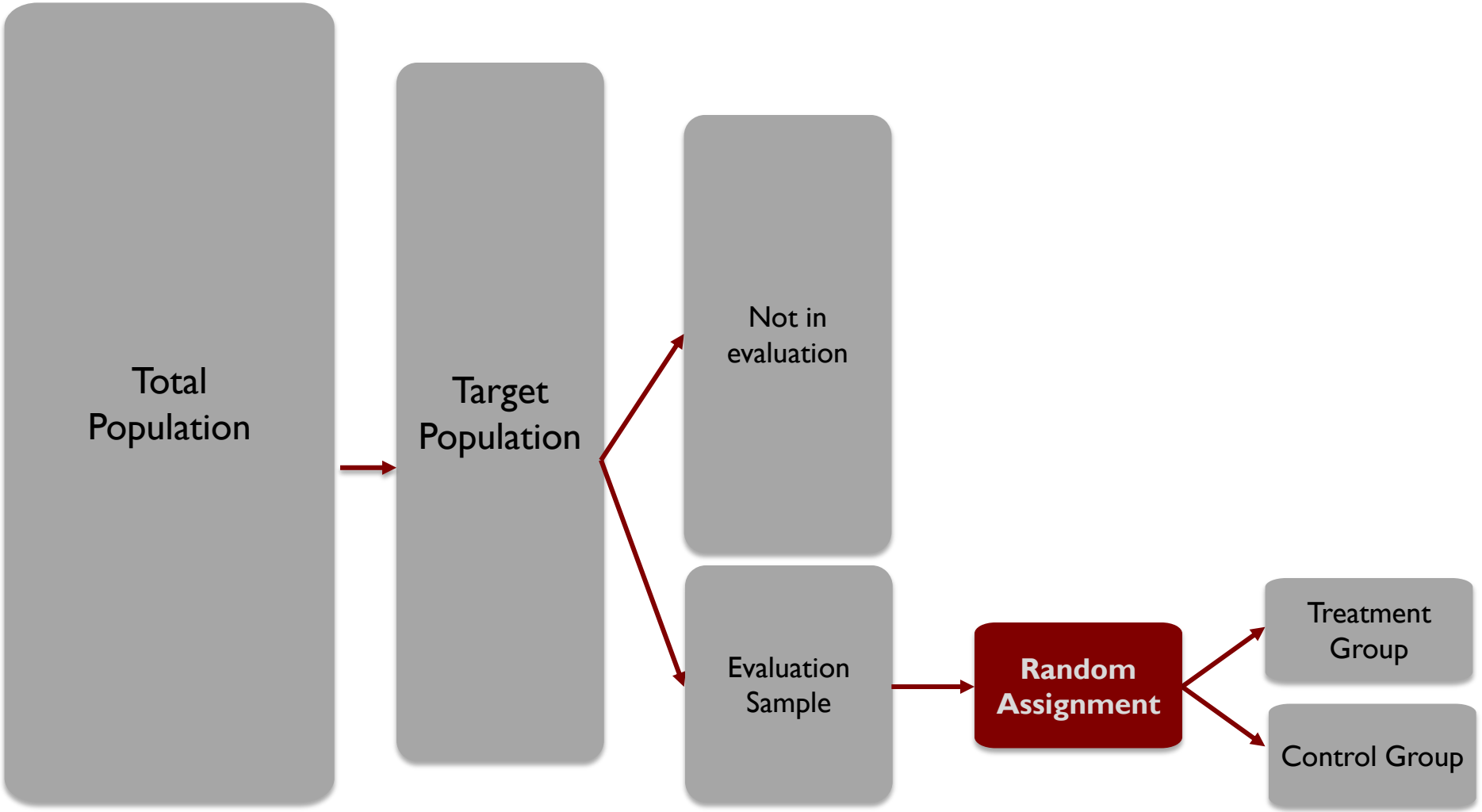
If properly designed and conducted, randomized experiments provide the most credible method to estimate the impact of a program

Testing Assumptions: Randomized Evaluations

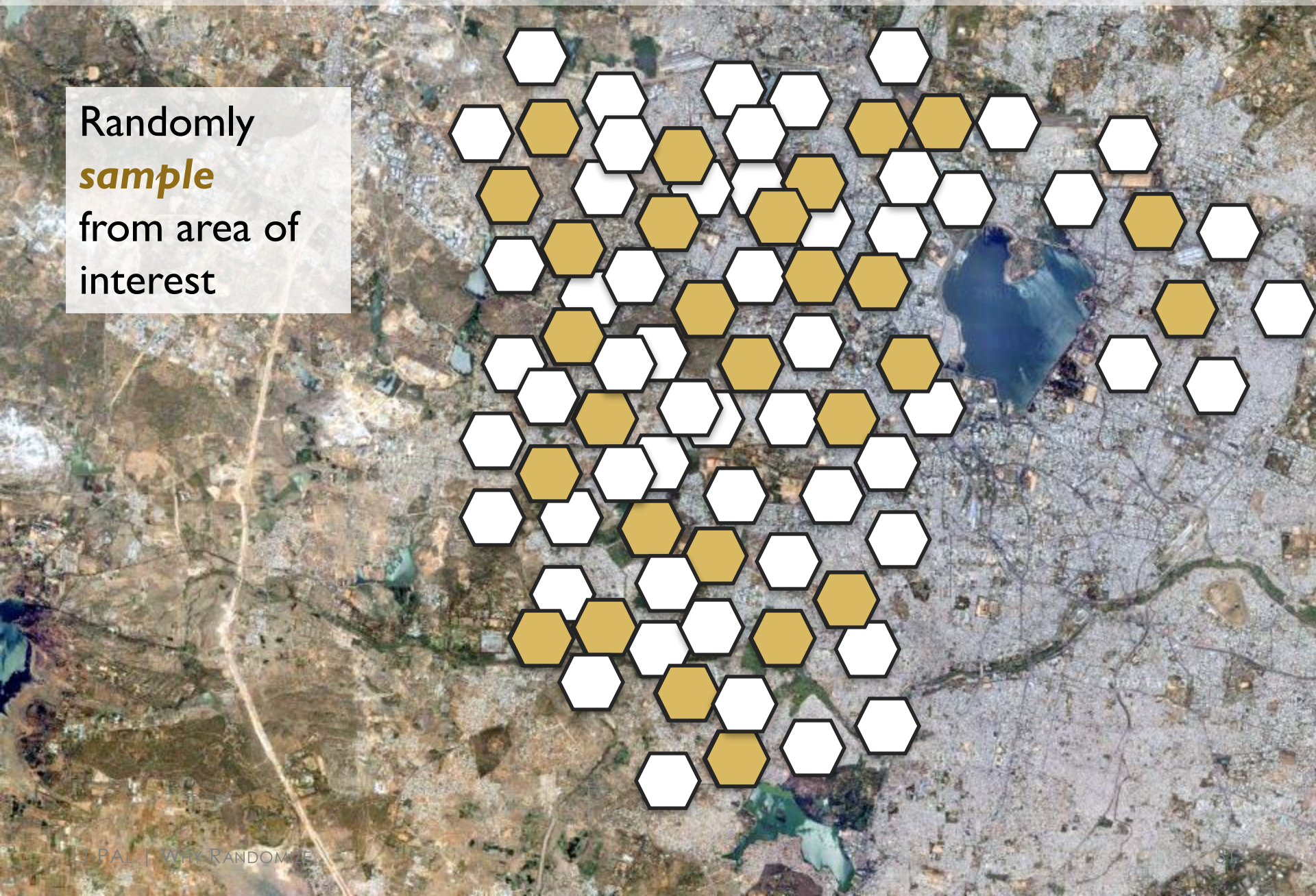
- What is the main assumption of randomized evaluation that must hold for it to give the true impact of the program?
  - No randomization failure: that randomization generates two statistically identical groups

- How can you test whether this assumption is true?
  - Balance test – compare their characteristics at baseline (beginning of the program)

Basic set-up of a Randomized Evaluation



Random Sampling and Random Assignment

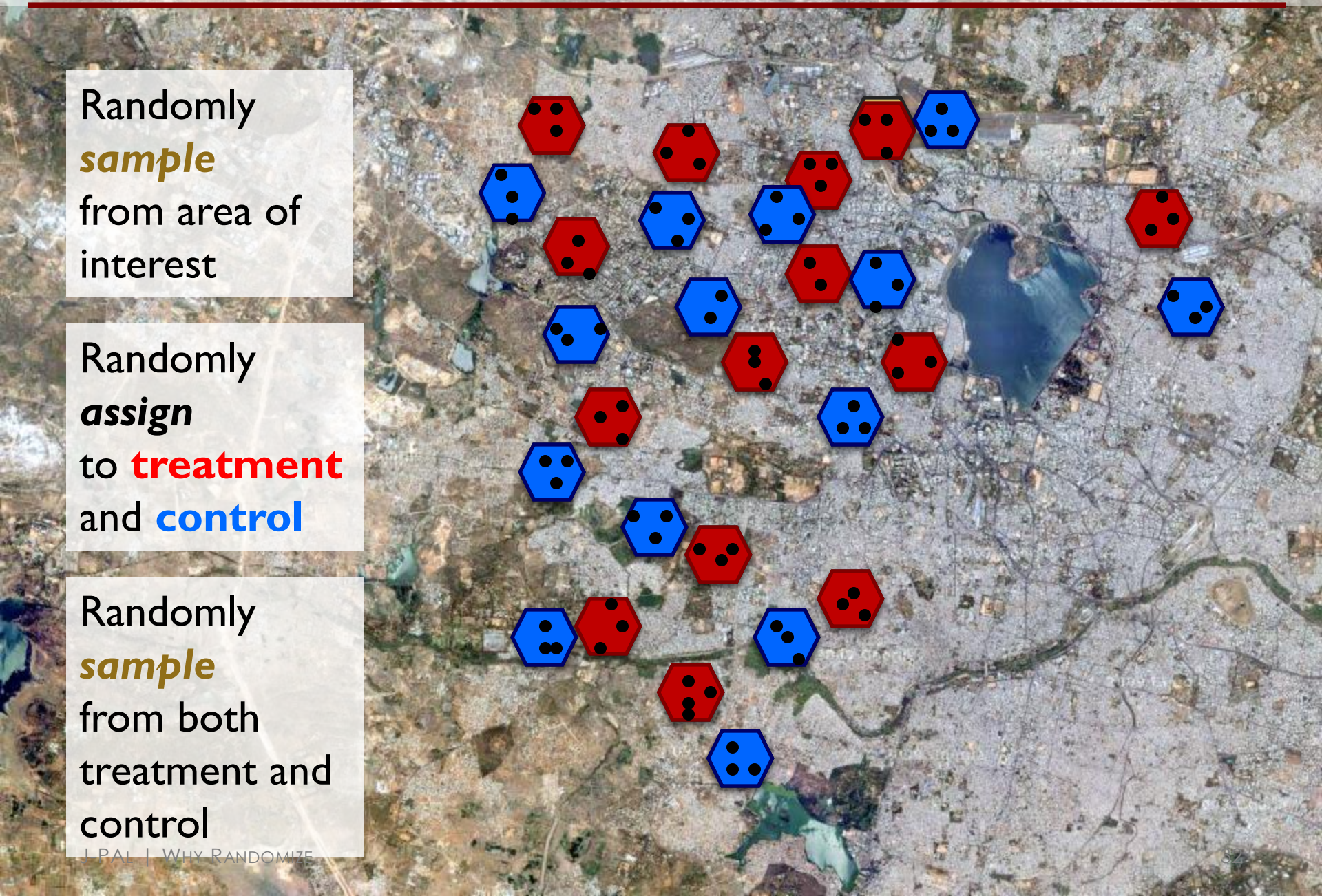


Random Sampling and Random Assignment

Randomly ***sample*** from area of interest

Randomly ***assign*** to **treatment** and **control**

Randomly ***sample*** from both treatment and control



Non or Quasi-Experimental Methods

- Let us look at other methods of estimating impact using the data from the schools that got a balsakhi
    1. Pre – Post (Before vs. After)
    2. Simple difference
    3. Difference-in-difference

- Other methods can be effective if the specific conditions needed for that method's assumption to hold exist

- Limitation: Conditions needed for them to be valid do not always apply

1 - Pre-post (Before vs. After)

- Look at average change in test scores over the school year for the balsakhi children



1 - Pre-post (Before vs. After)

Average post-test score for children with a balsakhi	51.22
Average pretest score for children with a balsakhi	24.80
Difference	26.42

QUESTION: Under what conditions can this difference (26.42) be interpreted as the impact of the balsakhi program?

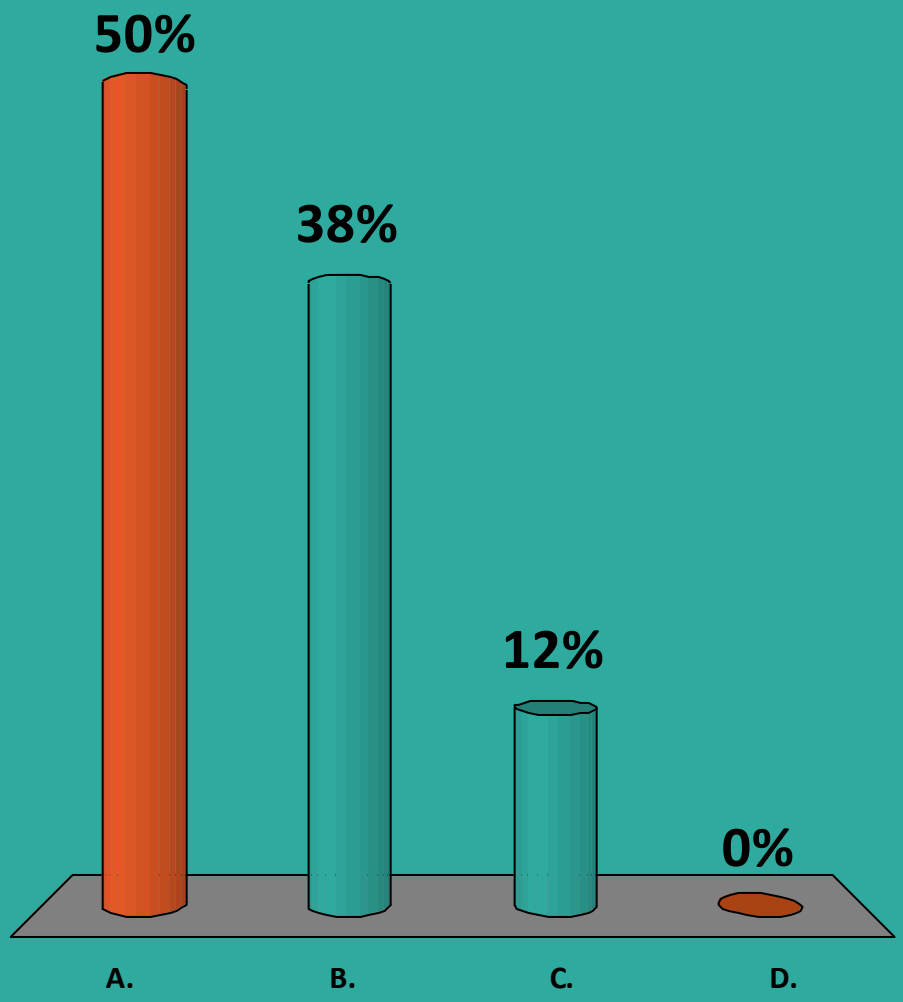
Which of the following represents the counterfactual in this case:

A. Balsakhi students before participating in the program

B. The non-Balsakhi students in the same schools

C. Students from other schools in Vadodara where the Balsakhi program is not being implemented

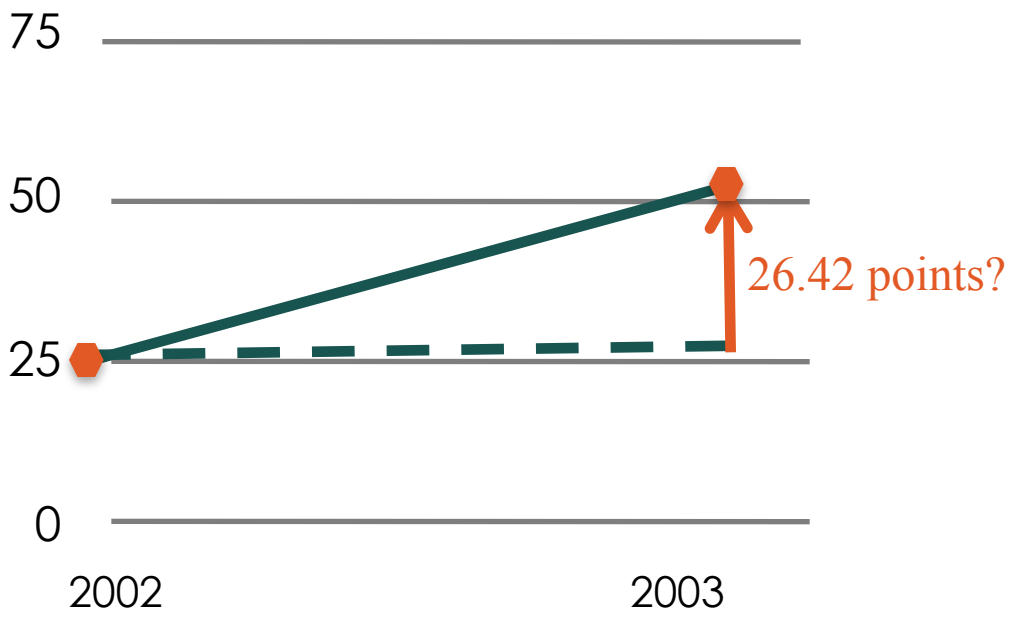
D. None of the above



What would have happened without Balsakhi?

Method 1: Before vs. After

Impact = 26.42 points?



2 - Simple difference

Compare test scores of…



Children who got balsakhi

With test scores of…



Children who did not get balsakhi

2 - Simple difference

Average score for children with a balsakhi	**51.22**
Average score for children without a balsakhi	56.27
Difference	-5.05

QUESTION: Under what conditions can this difference (-5.05) be interpreted as the impact of the balsakhi program?

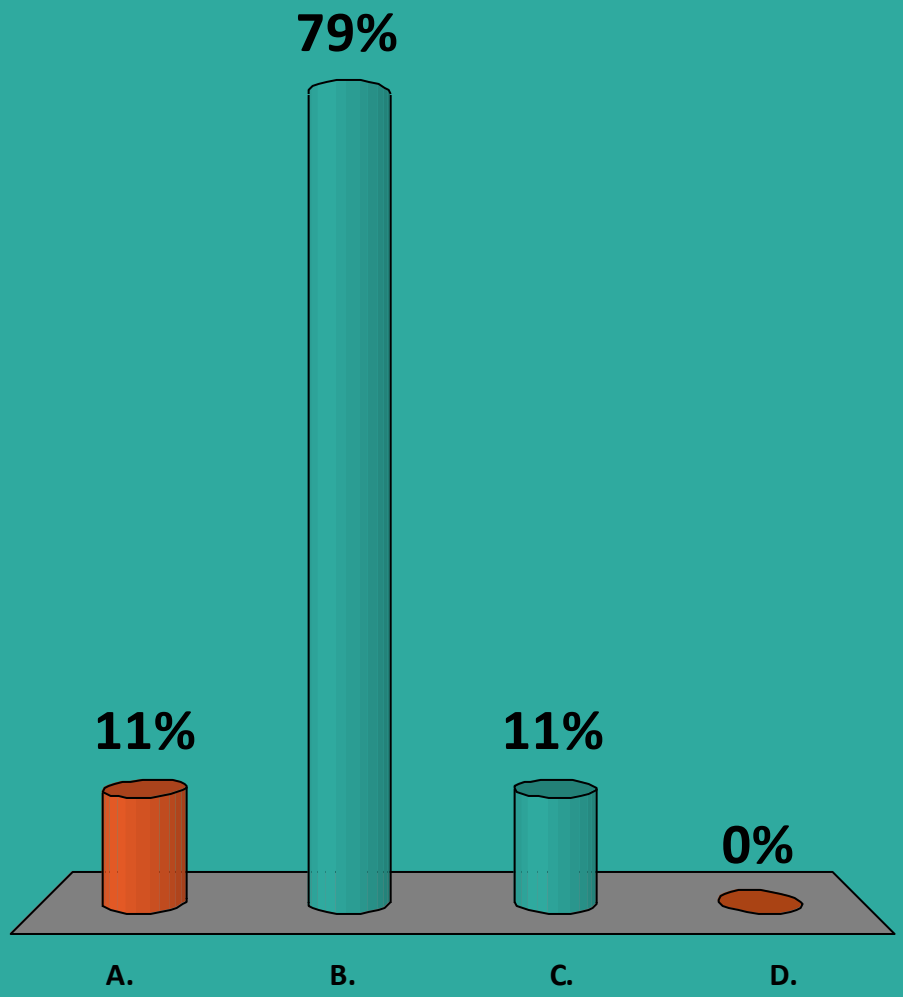
Which of the following represents the counterfactual in this case:

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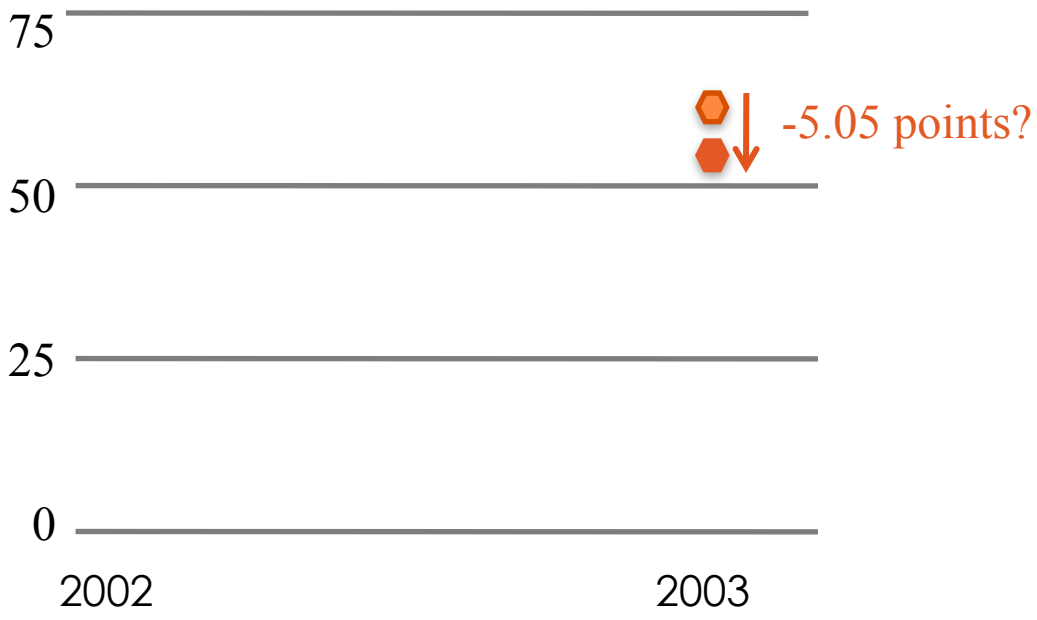
D. None of the above



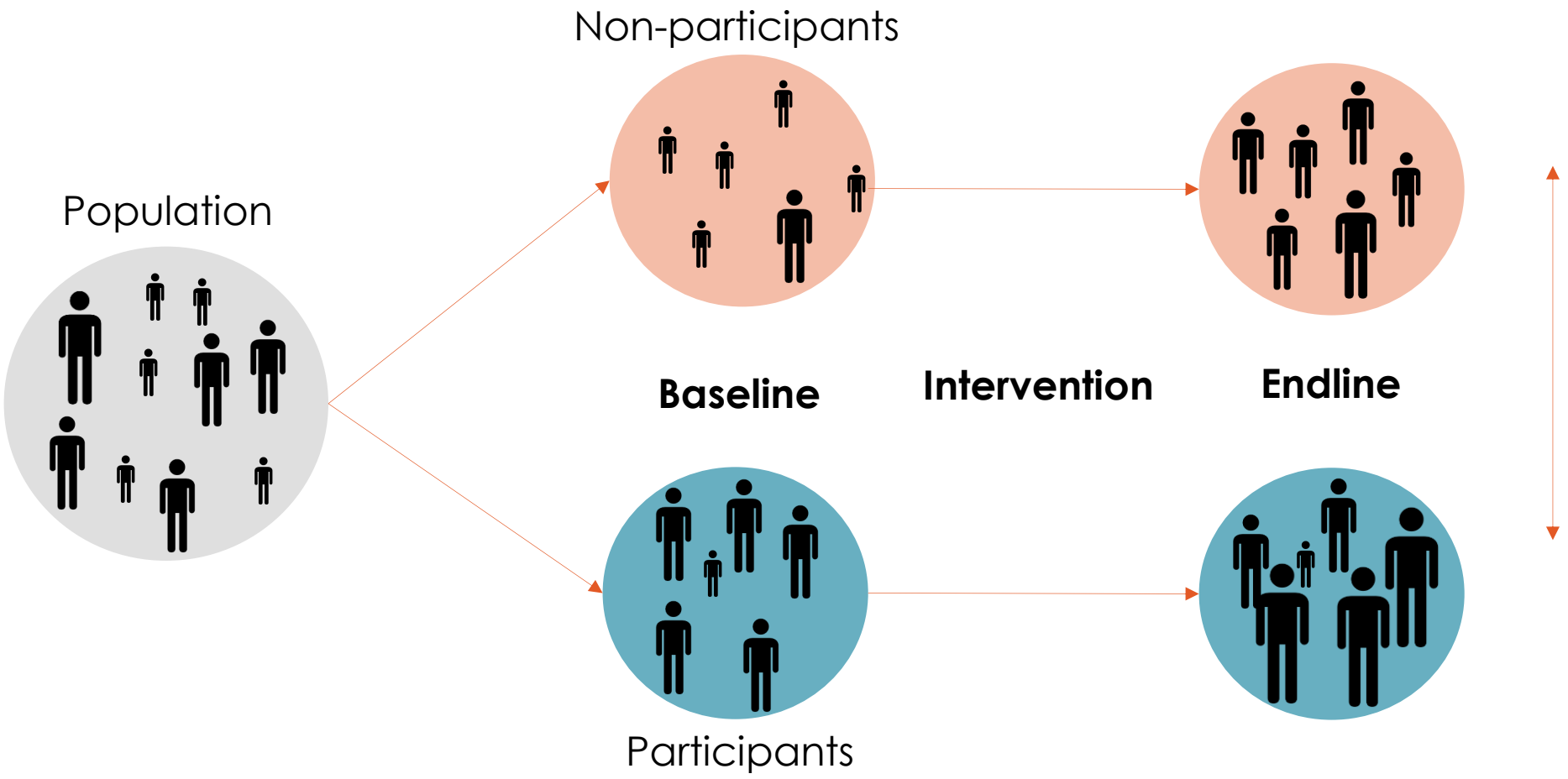
What would have happened without balsakhi?

Method 2: Simple Comparison

Impact = -5.05 points?



Selection Bias



Is this difference due to the program?

Or pre-existing differences?

3 – Difference-in-Differences

Compare gains in test scores of…



With gains in test scores of…



Children who got balsakhi

Children who did not get balsakhi

3 – Difference-in-difference

	Pretest	Post-test	Difference
Average score for children with a balsakhi	24.80	51.22	26.42

3 – Difference-in-difference

	Pretest	Post-test	Difference
Average score for children with a balsakhi	24.80	51.22	26.42
Average score for children without a balsakhi	36.67	56.27	19.60

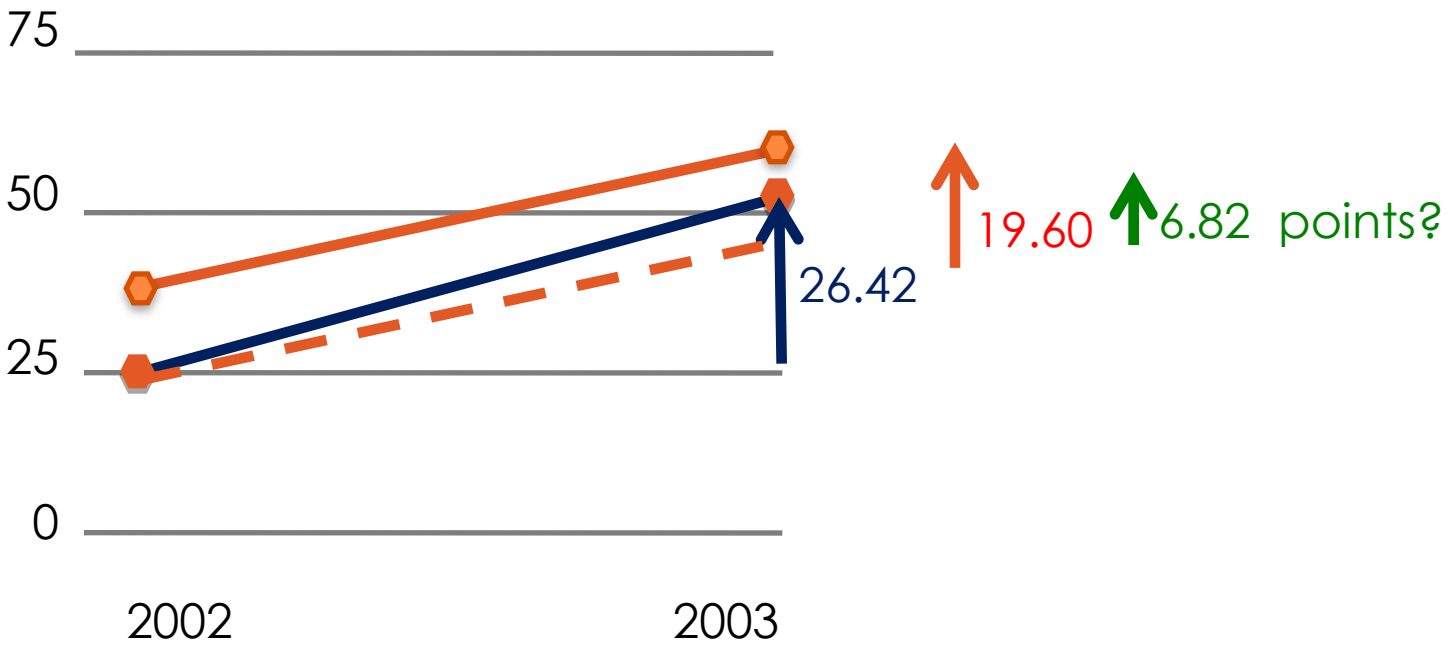
3 – Difference-in-difference

	Pretest	Post-test	Difference
Average score for children with a balsakhi	24.80	51.22	26.42
Average score for children without a balsakhi	36.67	56.27	19.60
Difference			6.82

QUESTION: Under what conditions can this difference (6.82) be interpreted as the impact of the balsakhi program?

What would have happened without balsakhi?

- Method 3: Difference-in-differences



4 – Multivariate Regression

A regression creates a model that estimates the relationship between a dependent and independent variable.

$$\text{Literacy} = a + b_1(\text{Tutor}) + \mathbf{b_2(Controls)} + \text{Error}$$

We can control for observable factors like:

- pretest score
- income
- gender
- age

	Difference
Average score for children with a balsakhi	1.92

We can't control for unobservable characteristics. This bias is called omitted variable bias.

5 – Other Methods

- There are more sophisticated non-experimental methods to estimate programme impacts:
  - Matching
  - Instrumental Variables
  - Regression Discontinuity

- These methods rely on being able to "mimic" the counterfactual under certain assumptions

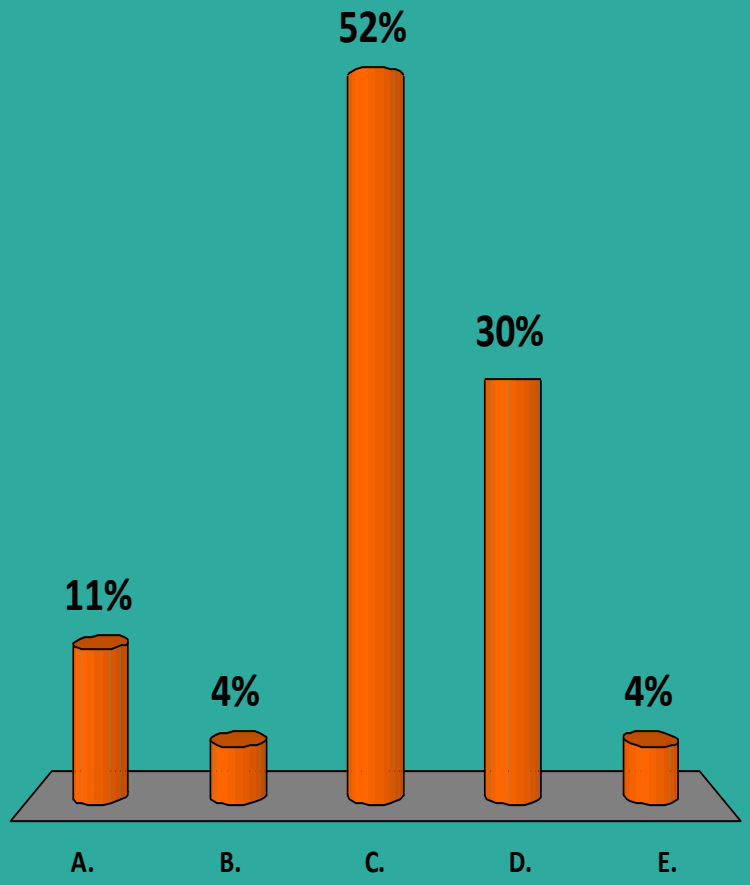
- Problem: Assumptions are not testable

Which of these methods do you think is closest to the truth?

Method	Impact Estimate
(1) Pre-post	26.42*
(2) Simple Difference	-5.05*
(3) Difference-in-Difference	6.82*
(4) Regression	1.92

*: Statistically significant at the 5% level

A. Pre-Post
B. Simple Difference
C. Difference-in-Differences
D. Regression
E. Don't know



Impact of Balsakhi - Summary

Method	Impact Estimate
(1) Pre-Post	26.42*
(2) Simple Difference	-5.05*
(3) Difference-in-Differences	6.82*
(4) Regression	1.92
(5) Randomized Experiment	**5.87***

*: Statistically significant at the 5% level

Bottom Line: Which method we use matters!

IV – CONCLUSIONS

Conclusions - Why Randomize?

- There are many ways to estimate a program's impact
- This course argues in favor of one: randomized experiments
  - Conceptual argument: If properly designed and conducted, randomized experiments provide the most credible method to estimate the impact of a program
  - Empirical argument: Different methods can generate different impact estimates

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Thank you!